

**Haramaya University  
School of Agricultural Economics & Agribusiness**

**ECONOMTERICS (AgEc 312)**

**For AgEc & ABVM Program**

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## **Econometrics**

### **Econometrics**

#### **1.1. Background**

Econometrics is about how we can use economic or social science theory and data, along with tools from statistics to answer how much type questions. It integrates mathematical knowledge, statistical skills and economic theories to solve business and economic problems of agribusiness firms. The economic theories we learn in various economics courses suggest many relationships among economic variables. For instance, in microeconomics we learn demand and supply models in which the quantities demanded and supplied of a good depend on its price. In macroeconomics, we study investment function to explain the amount of aggregate investment in the economy as the rate of interest changes; and consumption function that relates aggregate consumption to the level of aggregate disposable income. Each of such specifications involves a relationship among economic variables. As economists, we may be interested in questions such as: If one variable changes in a certain magnitude, by how much will another variable change? Also, given that we know the value of one variable; can we forecast or predict the corresponding value of another? The purpose of studying the relationships among economic variables and attempting to answer questions of the type raised here, is to help us understand the real economic world we live in. However, economic theories that postulate the relationships between economic variables have to be checked against data obtained from the real world. If empirical data verify the relationship proposed by economic theory, we accept the theory as valid. If the theory is incompatible with the observed behavior, we either reject the theory or in the light of the empirical evidence of the data, modify the theory. To provide a better understanding of economic relationships and a better guidance for economic policy making we also need to know the quantitative relationships between the different economic variables. We obtain these quantitative measurements taken from the real world. The field of knowledge which helps us to carryout such an evaluation of economic theories in empirical terms is econometrics.

#### **1.2. Learning Task Objectives**

This learning task is designed to equip students with the basic concepts and principles of econometrics. It attempts to foster students ability to estimate ordinary least square (OLS) regression; make predictions using regression analysis; integrate mathematics, statistics and economic theories to solve problems in agribusiness and value chain. In addition, it exposes the students to various statistical packages for analyzing data and provides them with practical activities to utilize the software packages in appropriate contexts or situations. Specific objectives of the learning task involve:

- Measure and analyze the association between economic variables
- Formulate models to represent and study economic relationships
- Estimate economic parameters important for policy formation and evaluation
- Forecast economic relationships

## **Econometrics**

### **1.3. Fundamental concepts of Econometrics**

#### **Pre-test**

1. What is econometrics?
2. What is the importance of econometric models?
3. Mention desirable properties of econometric models
4. Differentiate between economic and econometric model
5. What are the goals of econometrics?

#### **1.3.1. Definition and scope of econometrics**

What is Econometrics?

Simply stated Econometric means economic measurement. The metric part of the word signifies measurement and econometrics is concerned with the measuring of economic relationships.

It is a social science in which the tools of economic theory, mathematics and statistical inference are applied to the analysis of economic phenomena (Arthur Goldberger).

In the words of Maddala econometrics is the application of statistical and mathematical methods to the analysis of economic data, with a purpose of giving empirical content to economic theories and verifying them or refuting them.

Econometrics utilizes economic theory, as embodied in an econometric Model; facts, as summarized by relevant data, and statistical theory, as refined into econometric techniques to measure and to test empirically certain relationships among economic variables.

- It is a special type of economic analysis and research in which the general economic theory formulated in mathematical form (i.e. mathematical economics) is combined with empirical measurement (i.e. statistics) of economic phenomena.

#### **1.4. Why a Separate Discipline?**

As the definition suggests econometrics is an amalgam of economic theory, mathematical statistics and economic statistics. But: a distinction has to be made between Econometrics, and economic theory, statistics and mathematics.

1. Economic theory makes statements or hypotheses that are mostly of qualitative nature.

Example: Other things remaining constant (*ceteris paribus*) a reduction in the price of a commodity is expected to increase the quantity demanded. And Economic theory postulates an inverse relationship between price and quantity demanded of a commodity. But the theory does not provide numerical value as the measure of the relationship between the two. Here comes the task of the econometrician to provide the numerical value by which the quantity will go up or down as a result of changes in the price of the commodity.

2. Economic statistics is concerned with collecting, processing and presenting economic data (descriptive statistics).

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Example: collecting and refining data on national accounts, index numbers, employment, prices, etc.

3. Mathematical statistics and mathematical economics do provide much of the tools used in Econometrics. But Econometrics needs special methods to deal with economic data which are never experimental data.

Examples: Errors of measurement, problem of multicollinearity, problem of serial correlation are only econometric problems and are not concerns of mathematical statistics.

Econometrics utilizes these data to estimate quantitative economic relationships and to test hypothesis about them. The Econometrician is called upon to develop special methods of analysis and deal with such kinds of Econometric problems.

### 1.4.1. Economic models vs. Econometric models

A model is any representation of an actual phenomenon such as an actual system or process. The real world system is represented by the model in order to explain it, to predict it, and to control it. Any model represents a compromise between reality and manageability. A given representation of real world system can be a model if it fulfills the following requirements.

- (1) It must be a reasonable representation of the real world system and in that sense it should be realistic.
- (2) On the other hand it must be manageable in that it yields certain insights or conclusions.

A good model is both realistic and manageable. A highly realistic but too complicated model is a bad model in the sense it is not manageable. A model that is highly manageable but so idealized that it is unrealistic not accounting for important components of the real world system, is a bad model too. In general to find the proper balance between realism and manageability is the essence of good Modeling. Thus a good model should, on the one hand, specify the interrelationship among the parts of a system in a way that is sufficiently detailed and explicit and, on the other hand, it should be sufficiently simplified and manageable to ensure that the model can be readily analyzed and conclusions can be reached concerning the real world.

### Economic models

Any economic theory is an observation from the real world. For one reason, the immense complexity of the real world economy makes it impossible for us to understand all interrelationships at once. Another reason is that all the interrelationships are not equally important for the understanding of the economic phenomenon under study. The sensible procedure is therefore, to pick up the important factors and relationships relevant to our problem and to focus our attention on these alone. Such a deliberately simplified analytical framework is called on economic model. It is an organized set of relationships that describes the functioning of an economic entity under a set of simplifying assumptions. All economic

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reasoning is ultimately based on models. Economic models consist of the following three basic structural elements.

1. A set of variables
2. A list of fundamental relationships and
3. A number of strategic coefficients

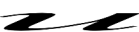
### Econometric models

The most important characteristic of economic relationships is that they contain a random element which is ignored by mathematical economic models which postulate exact relationships between economic variables.

Example: Economic theory postulates that the demand for a commodity depends on its price, on the prices of other related commodities, on consumers' income and on tastes. This is an exact relationship which can be written mathematically as:

$$Q = b_0 + b_1P + b_2P_0 + b_3Y + b_4t$$

The above demand equation is exact. However, many more factors may affect demand. In econometrics the influence of these other factors is taken into account by the introducing random variable. In our example, the demand function studied with the tools of econometrics would be of the stochastic form:

$Q = b_0 + b_1P + b_2P_0 + b_3Y + b_4t + u$ , where  stands for the random factors which affect the quantity demanded. The random term (also called *error term* or *disturbance term*) is a surrogate variable for important variables excluded from the model, errors committed and measurement errors.

### 1.5. Methodology of Econometrics

Econometric research is concerned with the measurement of the parameters of economic relationships and with the predication of the values of economic variables. The relationships of economic theory which can be measured with econometric techniques are relationships in which some variables are postulated as causes of the variation of other variables. Starting with the postulated theoretical relationships among economic variables, econometric research or inquiry generally proceeds along the following lines/stages.

1. Specification of the model
2. Estimation of the model
3. Evaluation of the estimates
4. Evaluation of the forecasting power of the estimated model

#### 1. Specification of the model

In this step the econometrician has to express the relationships between economic variables in mathematical form. The step involves the determination of three important issues:

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- Determine dependent and independent (explanatory) variables to be included in the model,
- Determine a priori theoretical expectations about the size and sign of the parameters of the function, and
- Determine mathematical form of the model (number of equations, specific form of the equations, etc).

Specification of the econometric model will be based on economic theory and on any available information related to the phenomena under investigation. Thus, specification of the econometric model presupposes knowledge of economic theory and familiarity with the particular phenomenon being studied. Specification of the model is the most important and the most difficult stage of any econometric research. It is often the weakest point of most econometric applications. In this stage there exists enormous degree of likelihood of committing errors or incorrectly specifying the model. The most common errors of specification are:

- a. Omissions of some important variables from the function.
- b. The omissions of some equations (for example, in simultaneous equations model).
- c. The mistaken mathematical form of the functions.

Such misspecification errors may associate to one or more reasons. Some of the common reasons for incorrect specification of the econometric models are:

- imperfections, looseness of statements in economic theories
- limited knowledge of the factors which are operative in any particular case
- formidable obstacles presented by data requirements in the estimation of large models

### **2. Estimation of the model**

This is purely a technical stage which requires knowledge of the various econometric methods, their assumptions and the economic implications for the estimates of the parameters. This stage includes the following activities.

- i. Gathering of the data on the variables included in the model.
- ii. Examination of the identification conditions of the function (especially for simultaneous equations models).
- iii. Examination of the aggregations problems involved in the variables of the function.
- iv. Examination of the degree of correlation between the explanatory variables (i.e. examination of the problem of multicollinearity).
- v. Choice of appropriate economic techniques for estimation, i.e. to decide a specific econometric method to be applied in estimation; such as, OLS, MLM

### **3. Evaluation of the estimates**

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This stage consists of deciding whether the estimates of the parameters are theoretically meaningful and statistically significant. This stage enables the econometrician to evaluate the results of calculations and determine the reliability of the results. For this purpose we use various criteria which may be classified into three groups:

- i. Economic a priori criteria: These criteria are determined by economic theory and refer to the size and sign of the parameters of economic relationships.
- ii. Statistical criteria (first-order tests): These are determined by statistical theory and aim at the evaluation of the statistical reliability of the estimates of the parameters of the model. Correlation coefficient test, standard error test, t-test, F-test, and  $R^2$ -test are some of the most commonly used statistical tests.
- iii. Econometric criteria (second-order tests): These are set by the theory of econometrics and aim at the investigation of whether the assumptions of the econometric method employed are satisfied or not in any particular case. The econometric criteria serve as a second order test (as test of the statistical tests) i.e. they determine the reliability of the statistical criteria; they help us establish whether the estimates have the desirable properties of unbiasedness, consistency, etc. Econometric criteria aim at the detection of the violation or validity of the assumptions of the various econometric techniques.

### 4) Evaluation of the forecasting power of the model

Forecasting is one of the aims of econometric research. However, before using an estimated model for forecasting by some way or another, the predictive power and other requirements of the model need to be checked. It is possible that the model may be economically meaningful and statistically and econometrically correct for the sample period for which the model has been estimated. Yet it may not be suitable for forecasting due to various factors (reasons). Therefore, this stage involves the investigation of the stability of the estimates and their sensitivity to changes in the size of the sample. Consequently, we must establish whether the estimated function performs adequately outside the sample data which require model performance test under extra sample.

### 1.6. Desirable Properties of an Econometric Model

An econometric model is a model whose parameters have been estimated with some appropriate econometric technique. The goodness of an econometric model is judged customarily based on the following desirable properties.

1. Theoretical plausibility: The model should be compatible with the postulates of economic theory and adequately describe the economic phenomena to which it relates.
2. Explanatory ability: The model should be able to explain the observations of the actual world. It must be consistent with the observed behaviour of the economic variables whose relationship it determines.
3. Accuracy of the estimates of the parameter: The estimates of the coefficients should be accurate in the sense that they should approximate as best as possible the true

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parameters of the structural model. The estimates should if possible possess the desirable properties of unbiasedness, consistency and efficiency.

4. Forecasting ability: The model should produce satisfactory predictions of future values of the dependent (endogenous) variables.
5. Simplicity: The model should represent the economic relationships with maximum simplicity. The fewer the equations and the simpler their mathematical form, the better the model provided that the other desirable properties are not affected by the simplifications of the model.

### 1.7. Goals of Econometrics

Basically there are three main goals of Econometrics. They are:

- i) Analysis i.e. testing economic theory
- ii) Policy making i.e. obtaining numerical estimates of the coefficients of economic relationships for policy simulations.
- iii) Forecasting i.e. using the numerical estimates of the coefficients in order to forecast the future values of economic magnitudes.

|                         |
|-------------------------|
| <b>Review questions</b> |
|-------------------------|

- |                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ol style="list-style-type: none"><li>1. How would you define econometrics?</li><li>2. How does it differ from mathematical economics and statistics?</li><li>3. Describe the main steps involved in any econometrics research.</li><li>4. Differentiate between economic and econometric model.</li><li>5. What are the goals of econometrics?</li></ol> |
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## CHAPTER 2

### CORRELATION THEORY

#### Pre-test Questions

1. How do you define correlation?
2. Which correlation measurement methods do you know?

#### 2.1. Correlation Analysis

Economic variables have a great tendency of moving together and very often data are given in pairs of observations in which there is a possibility that the change in one variable is on average accompanied by the change of the other variable. This situation is known as correlation.

Correlation may be defined as the degree of relationship existing between two or more variables. The degree of relationship existing between two variables is called simple correlation. The degree of relationship connecting three or more variables is called multiple correlations. In this unit, we shall examine only simple correlation. A correlation is also said to be partial if it studies the degree of relationship between two variables keeping all other variables connected with these two are constant.

Correlation may be linear, when all points (X, Y) on scatter diagram seem to cluster near a straight, or nonlinear, when all points seem to lie near a curve. In other words, correlation is said to be linear if the change in one variable brings a constant change of the other. It may be non-linear if the change in one variable brings a different change in the other.

Correlation may also be positive or negative. Correlation is said to positive if an increase or a decrease in one variable is accompanied by an increase or a decrease by the other in which both variables are changed with the same direction. For example, the correlation between price of a commodity and its quantity supplied is positive since as price rises, quantity supplied will be increased and vice versa. Correlation is said to negative if an increase or a decrease in one variable is accompanied by a decrease or an increase in the other in which both are changed with opposite direction. For example, the correlation between price of a commodity and its quantity demanded is negative since as price rises, quantity demanded will be decreased and vice versa.

#### 2.2. Methods of Measuring Correlation

In correlation analysis there are two important things to be addressed. These are the type of co-variation existed between variables and its strength. And the types of correlation mentioned before do not show to us the strength of co-variation between variables. There are three methods of measuring correlation. These are:

1. The Scattered Diagram or Graphic Method
2. The Simple Linear Correlation coefficient

## 3. The coefficient of Rank Correlation

### The Scattered Diagram or Graphic Method

The scatter diagram is a rectangular diagram which can help us in visualizing the relationship between two phenomena. It puts the data into X-Y plane by moving from the lowest data set to the highest data set. It is a non-mathematical method of measuring the degree of co-variation between two variables. Scatter plots usually consist of a large body of data. The closer the data points come together and make a straight line, the higher the correlation between the two variables, or the stronger the relationship.

If the data points make a straight line going from the origin out to high x- and y-values, then the variables are said to have a positive correlation. If the line goes from a high-value on the y-axis down to a high-value on the x-axis, the variables have a negative correlation.

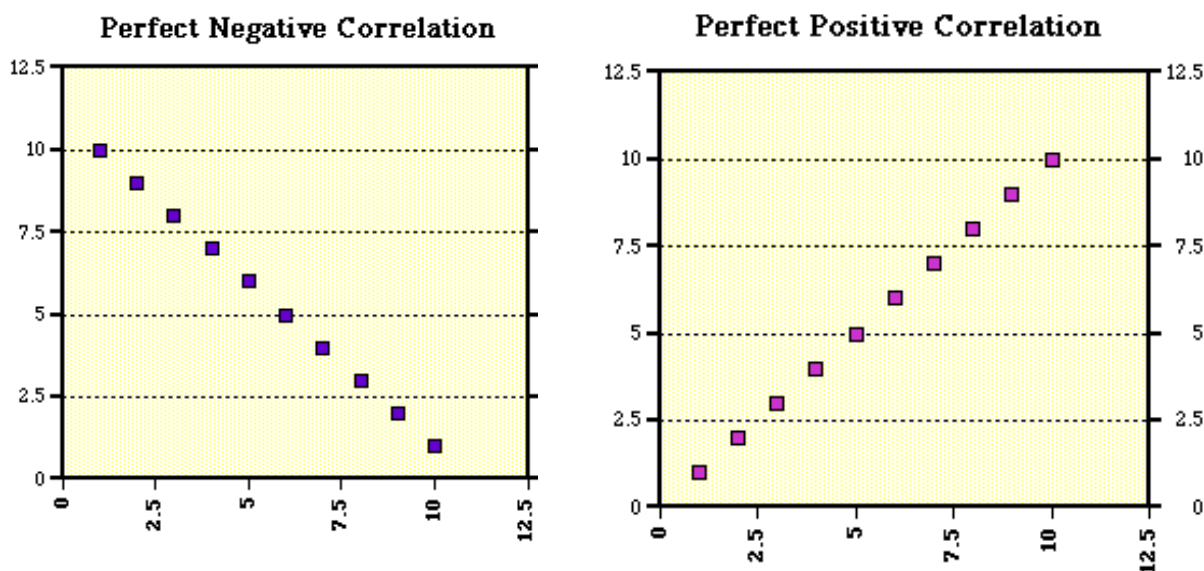


Figure 1: Perfect linear correlations

A perfect positive correlation is given the value of 1. A perfect negative correlation is given the value of -1. If there is absolutely no correlation present the value given is 0. The closer the number is to 1 or -1, the stronger the correlation, or the stronger the relationship between the variables. The closer the number is to 0, the weaker the correlation.

Two variables may have a positive correlation, negative correlation, or they may be uncorrelated. This holds true both for linear and nonlinear correlation. Two variables are said to be positively correlated if they tend to change together in the same direction, that is, if they tend to increase or decrease together. Such positive correlation is postulated by economic theory for the quantity of a commodity supplied and its price. When the price increases the quantity supplied increases. Conversely, when price falls the quantity supplied decreases. Negative correlation: Two variables are said to be negatively correlated if they tend to change in the opposite direction: when X increases Y decreases, and vice versa. For example, saving

and household size are negatively correlated. When price increases, demand for the commodity decreases and when price falls demand increases.

## The Population Correlation Coefficient $\rho$ and its Sample Estimate $r$

In the light of the above discussions it appears clear that we can determine the kind of correlation between two variables by direct observation of the scatter diagram. In addition, the scatter diagram indicates the strength of the relationship between the two variables. This section is about how to determine the type and degree of correlation using a numerical result. For a precise quantitative measurement of the degree of correlation between Y and X we use a parameter which is called the correlation coefficient and is usually designated by the Greek letter  $\rho$ . Having as subscripts the variables whose correlation it measures,  $\rho$  refers to the correlation of all the values of the population of X and Y. Its estimate from any particular sample (the sample statistic for correlation) is denoted by  $r$  with the relevant subscripts. For example if we measure the correlation between X and Y the population correlation coefficient is represented by  $\rho_{xy}$  and its sample estimate by  $r_{xy}$ . The simple correlation coefficient is used to measure relationships which are simple and linear only. It cannot help us in measuring non-linear as well as multiple correlations. Sample correlation coefficient is defined by the formula

$$r = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{\sqrt{n \sum x_i^2 - (\sum x_i)^2} \sqrt{n \sum y_i^2 - (\sum y_i)^2}} \quad 2.1$$

$$\text{Or } r_{xy} = \frac{\sum x_i y_i}{\sqrt{\sum x_i^2} \sqrt{\sum y_i^2}} \quad 2.2$$

Where,



We will use a simple example from the theory of supply. Economic theory suggests that the quantity of a commodity supplied in the market depends on its price, *ceteris paribus*. When price increases the quantity supplied increases, and vice versa. When the market price falls producers offer smaller quantities of their commodity for sale. In other words, economic theory postulates that price (X) and quantity supplied (Y) are positively correlated.

Example 2.1: The following table shows the quantity supplied for a commodity with the corresponding price values. Determine the type of correlation that exists between these two variables.

Table 1: Data for computation of correlation coefficient

| Time period(in days) | Quantity supplied $Y_i$ (in tons) | Price $X_i$ (in shillings) |
|----------------------|-----------------------------------|----------------------------|
| 1                    | 10                                | 2                          |
| 2                    | 20                                | 4                          |
| 3                    | 50                                | 6                          |
| 4                    | 40                                | 8                          |
| 5                    | 50                                | 10                         |
| 6                    | 60                                | 12                         |
| 7                    | 80                                | 14                         |
| 8                    | 90                                | 16                         |
| 9                    | 90                                | 18                         |
| 10                   | 120                               | 20                         |

To estimate the correlation coefficient we, compute the following results.

Table 2: Computations of inputs for correlation coefficients

| Y       | X   | $x_i = Y_i - \bar{Y}$ | $y_i = Y_i - \bar{Y}$ | $x^2$ | $y^2$ | $x_i y_i$ | XY   | $X^2$ | $Y^2$ |
|---------|-----|-----------------------|-----------------------|-------|-------|-----------|------|-------|-------|
| 10      | 2   | -9                    | -51                   | 81    | 2601  | 459       | 20   | 4     | 100   |
| 20      | 4   | -7                    | -41                   | 49    | 1681  | 287       | 80   | 16    | 400   |
| 50      | 6   | -5                    | -11                   | 25    | 121   | 55        | 300  | 36    | 2500  |
| 40      | 8   | -3                    | -21                   | 9     | 441   | 63        | 320  | 64    | 1600  |
| 50      | 10  | -1                    | -11                   | 1     | 121   | 11        | 500  | 100   | 2500  |
| 60      | 12  | 1                     | -1                    | 1     | 1     | -1        | 720  | 144   | 3600  |
| 80      | 14  | 3                     | 19                    | 9     | 361   | 57        | 1120 | 196   | 6400  |
| 90      | 16  | 5                     | 29                    | 25    | 841   | 145       | 1440 | 256   | 8100  |
| 90      | 18  | 7                     | 29                    | 49    | 841   | 203       | 1620 | 324   | 8100  |
| 120     | 20  | 9                     | 59                    | 81    | 3481  | 531       | 2400 | 400   | 14400 |
| Sum=610 | 110 | 0                     | 0                     | 330   | 10490 | 1810      | 8520 | 1540  | 47700 |
| Mean=61 | 11  |                       |                       |       |       |           |      |       |       |

$$r = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sqrt{\sum x^2 - \frac{(\sum x)^2}{n}} \sqrt{\sum y^2 - \frac{(\sum y)^2}{n}}} = \frac{1810}{\sqrt{330} \sqrt{10490}} = 0.975$$

Or using the deviation form (Equation 2.2), the correlation coefficient can be computed as:

$$r = \frac{1810}{\sqrt{330} \sqrt{10490}} = 0.975$$

This result shows that there is a strong positive correlation between the quantity supplied and the price of the commodity under consideration.

The simple correlation coefficient has the value always ranging between -1 and +1. That means the value of correlation coefficient cannot be less than -1 and cannot be greater than +1. Its minimum value is -1 and its maximum value is +1. If  $r = -1$ , there is perfect negative correlation between the variables. If  $0 < r \leq +1$ , there is positive correlation between the two variables and movement from zero to positive one increases the degree of positive correlation. If  $r = +1$ , there is perfect positive correlation between the two variables. If the correlation coefficient is zero, it indicates that there is no linear relationship between the two variables. If the two variables are independent, the value of correlation coefficient is zero but zero correlation coefficient does not show us that the two variables are independent.

### Properties of Simple Correlation Coefficient

The simple correlation coefficient has the following important properties:

1. The value of correlation coefficient always ranges between -1 and +1.
2. The correlation coefficient is symmetric. That means

$$r_{xy} = r_{yx}, \quad \text{where,}$$

$r_{xy}$  is the correlation coefficient of X on Y and

$r_{yx}$  is the correlation coefficient of Y on X.

3. The correlation coefficient is independent of change of origin and change of scale. By change of origin we mean subtracting or adding a constant from or to every values of a variable and change of scale we mean multiplying or dividing every value of a variable by a constant.
4. If X and Y variables are independent, the correlation coefficient is zero. But the converse is not true.
5. The correlation coefficient has the same sign with that of regression coefficients.<sup>11</sup>
6. The correlation coefficient is the geometric mean of two regression coefficients.

$$r = \sqrt{b_{yx} * b_{xy}}$$

Though, correlation coefficient is most popular in applied statistics and econometrics, it has its own limitations. The major limitations of the method are:

1. The correlation coefficient always assumes linear relationship regardless of the fact whether the assumption is true or not.
2. Great care must be exercised in interpreting the value of this coefficient as very often the coefficient is misinterpreted. For example, high correlation between lung cancer and smoking does not show us smoking causes lung cancer.
3. The value of the coefficient is unduly affected by the extreme values
5. The coefficient requires the quantitative measurement of both variables. If one of the two variables is not quantitatively measured, the coefficient cannot be computed.

### **The Rank Correlation Coefficient**

The formulae of the linear correlation coefficient developed in the previous section are based on the assumption that the variables involved are quantitative and that we have accurate data for their measurement. However, in many cases the variables may be qualitative (or binary variables) and hence cannot be measured numerically. For example, profession, education, preferences for particular brands, are such categorical variables. Furthermore, in many cases precise values of the variables may not be available, so that it is impossible to calculate the value of the correlation coefficient with the formulae developed in the preceding section. For such cases it is possible to use another statistic, the rank correlation coefficient (or spearman's correlation coefficient.). We rank the observations in a specific sequence for example in order of size, importance, etc., using the numbers 1, 2, 3, ..., n. In other words, we assign ranks to the data and measure relationship between their ranks instead of their actual numerical values. Hence, the name of the statistic is given as rank correlation coefficient. If two variables X and Y are ranked in such way that the values are ranked in ascending or descending order, the rank correlation coefficient may be computed by the formula

$$r' = 1 - \frac{6 \sum D^2}{n(n^2 - 1)} \quad 2.3$$

Where,

D = difference between ranks of corresponding pairs of X and Y

n = number of observations.

The values that r may assume range from + 1 to - 1.

Two points are of interest when applying the rank correlation coefficient. Firstly, it does not matter whether we rank the observations in ascending or descending order. However, we must use the same rule of ranking for both variables. Second if two (or more) observations have the same value we assign to them the mean rank. Let's use example to illustrate the application of the rank correlation coefficient.

Example 2.2: A market researcher asks experts to express their preference for twelve different brands of soap. Their replies are shown in the following table.

**Table 3: Example for rank correlation coefficient**

| Brands of soap | A | B  | C | D | E  | F  | G | H | I | J | K  | L |
|----------------|---|----|---|---|----|----|---|---|---|---|----|---|
| Person I       | 9 | 10 | 4 | 1 | 8  | 11 | 3 | 2 | 5 | 7 | 12 | 6 |
| Person II      | 7 | 8  | 3 | 1 | 10 | 12 | 2 | 6 | 5 | 4 | 11 | 9 |

The figures in this table are ranks but not quantities. We have to use the rank correlation coefficient to determine the type of association between the preferences of the two persons. This can be done as follows.

**Table 4: Computation for rank correlation coefficient**

| Brands of soap | A | B  | C | D | E  | F  | G | H  | I | J | K  | L  | Total |
|----------------|---|----|---|---|----|----|---|----|---|---|----|----|-------|
| Person I       | 9 | 10 | 4 | 1 | 8  | 11 | 3 | 2  | 5 | 7 | 12 | 6  |       |
| Person II      | 7 | 8  | 3 | 1 | 10 | 12 | 2 | 6  | 5 | 4 | 11 | 9  |       |
| $D_i$          | 2 | 2  | 1 | 0 | -2 | -1 | 1 | -4 | 0 | 3 | 1  | -3 |       |
| $D_i^2$        | 4 | 4  | 1 | 0 | 4  | 1  | 1 | 16 | 0 | 9 | 1  | 9  | 50    |

The rank correlation coefficient (using Equation 2.3)

$$r' = 1 - \frac{6 \sum D^2}{n(n^2 - 1)} = 1 - \frac{6(50)}{12(12^2 - 1)} = 0.827$$

This figure, 0.827, shows a marked similarity of preferences of the two persons for the various brands of soap.

### Limitations of the Theory of Linear Correlation

Correlation analysis has serious limitations as a technique for the study of economic relationships.

Firstly: The above formulae for  $r$  apply only when the relationship between the variables is linear. However two variables may be strongly connected with a nonlinear relationship.

It should be clear that zero correlation and statistical independence of two variables ( $X$  and  $Y$ ) are not the same thing. Zero correlation implies zero covariance of  $X$  and  $Y$  so that  $r=0$ . Statistical independence of  $x$  and  $y$  implies that the probability of  $x_i$  and  $y_i$  occurring simultaneously is the simple product of the individual probabilities

$$P(x \text{ and } y) = p(x) p(y)$$

Independent variables do have zero covariance and are uncorrelated: the linear correlation coefficient between two independent variables is equal to zero. However, zero linear correlation does not necessarily imply independence. In other words uncorrelated variables may be statistically dependent. For example if  $X$  and  $Y$  are related so that the observations fall on a circle or on a symmetrical parabola, the relationship is perfect but not linear. The variables are statistically dependent.

Secondly, the second limitation of the theory is that although the correlation coefficient is a measure of the co-variability of variables, it does not necessarily imply any functional relationship between the variables concerned. Correlation theory does not establish, and/ or prove any causal relationship between the variables. It seeks to discover a co-variation exists, but it does not suggest that variations in, say, Y are caused by variations in X, or vice versa. Knowledge of the value of  $r$ , alone, will not enable us to predict the value of Y from X. A high correlation between variables Y and X may describe any one of the following situations:

- (1) variation in X is the cause of variation in Y,
- (2) variation in Y is the cause of variation X,
- (3) Y and X are jointly dependent, or there is a two- way causation, that is to say Y is the cause of (is determined by) X, but also X is the cause of (is determined by) Y. For example in any market:  $q = f(p)$ , but also  $p = f(q)$ , therefore there is a two way causation between q and p, or in other words p and q are simultaneously determined.
- (4) there is another common factor (Z), that affects X and Y in such a way as to show a close relation between them. This often occurs in time series when two variables have strong time trends (i.e. grow over time). In this case we find a high correlation between Y and X, even though they happen to be causally independent,
- (5) The correlation between X and Y may be due to chance.

#### **Statistical Software Application Practices**

**Use the following data and calculate simple correlation coefficient using Excell spreadsheet, STATA and SPS soft wares.**

| ID | X | Y  |
|----|---|----|
| 1  | 8 | 10 |
| 2  | 7 | 8  |
| 3  | 3 | 2  |
| 4  | 5 | 6  |
| 5  | 7 | 9  |
| 6  | 2 | 2  |
| 7  | 4 | 5  |

## CHAPTER THREE

### THE SIMPLE REGRESSION MODEL

1. What is regression Analysis?
2. What do you know about OLS (ordinary Least Squares) method?

Economic theories are mainly concerned with the relationships among various economic variables. These relationships, when phrased in mathematical terms, can predict the effect of one variable on another. The functional relationships of these variables define the dependence of one variable upon the other variable (s) in the specific form. In this regard regression model is the most commonly used and appropriate technique of econometric analysis. Regression analysis refers to estimating functions showing the relationship between two or more variables and corresponding tests. This section introduces students with the concept of simple linear regression analysis. It includes estimating a simple linear function between two variables. We will restrict our discussion in this part only to two variables and deal with more variables in the next section.

#### Ordinary Least Square Method (OLS) and Classical Assumptions

There are two major ways of estimating regression functions. These are the (ordinary) least square (OLS) method and maximum likelihood (MLH) method. Both the methods are basically similar to their application in estimations that you may be aware of in statistics courses. The ordinary least square method is the easiest and the most commonly used method as opposed to the maximum likelihood (MLH) method which is limited by its assumptions. For instance, the MLH method is valid only for large sample as opposed to the OLS method which can be applied to smaller samples. Owing to this merit, our discussion mainly focuses on the ordinary least square (OLS).

The (Ordinary) least square (OLS) method of estimating parameters or regression function is about finding or estimating values of the parameters ( $\alpha$  and  $\beta$ ) of the simple linear regression function given below for which the errors or residuals are minimized. Thus, it is about minimizing the residuals or the errors.

$$Y_i = \alpha + \beta X_i + u_i \quad (3.1)$$

The above identity represents population regression function (to be estimated from total enumeration of data from the entire population). But, most of the time it is difficult to generate population data owing to several reasons; and most of the time we use sample data and we estimate sample regression function. Thus, we use the following sample regression function for the derivation of the parameters and related analysis.

$$y_i = \alpha + \beta x_i + u_i \quad (3.2)$$

Before discussing the details of the OLS estimation techniques, let's see the major conditions that are necessary for the validity of the analysis, interpretations and conclusions of the regression function. These conditions are known as classical assumptions. In fact most of these conditions can be checked and secured very easily.

### i) Classical Assumptions

For the validity of a regression function and its attributes the data we use or the terms related to our regression function should fulfill the following conditions known as classical assumptions.

1. The error terms  $U_i$  are randomly distributed or the disturbance terms are not correlated. This means that there is no systematic variation or relation among the value of the error terms ( $U_i$  and  $U_j$ ); Where  $i = 1, 2, 3, \dots, j = 1, 2, 3, \dots$  and  $i \neq j$ . This is represented by zero covariance among the error terms summarized as follows:

$\text{Cov}(U_i, U_j) = 0$  for  $i \neq j$ . Note that the same argument holds for residual terms when we use sample data or sample regression function. Thus,  $\text{Cov}(e_i, e_j) = 0$  for



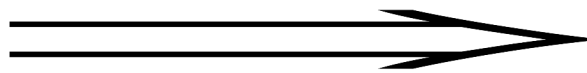
. Otherwise, the error terms do not serve an adjustment purpose rather it causes an autocorrelation problem.

2. The disturbance terms  $U_i$  have zero mean. This implies the sum of the individual disturbance terms is zero. The deviations of the values of some of the disturbance terms are negative; some are zero and some are positive and the sum or the average is zero. This is given by the following identities.

$$E(U_i) =$$

$$\frac{\sum U_i}{n} = 0$$

. Multiplying both sides by (sample size  $n$ ) we obtain the following.



$$E(U_i) =$$

$$\sum U_i = 0$$

. The same argument is true for sample regression function and so for residual terms given as

follows

$$\sum e_i$$

If this condition is not met, then the position of the regression function (or curve) will not be the same as where it is supposed to be. This results in an upward (if the mean of the error term or residual term is positive) or down ward (if the mean of the error term or residual term is negative) shift in the regression function. For instance, suppose we have the following regression function.

$$Y_i = \beta_0 + \beta_1 X_i + U_i$$

$$E(Y_i) = E(Y_i / X_i) = E(\beta_0 + \beta_1 X_i) + E(U_i)$$

$$E(Y_i) = E(Y_i / X_i) = \beta_0 + \beta_1 X_i \quad \text{if } E(U_i) = 0$$

Otherwise the estimated models will be biased and cause the regression function to shift. For instance, if  $E(U_i) > 0$  (or positive) it is going to shift the estimation upward from the true representative model. Similar argument is true for residual term of sample regression function. This is demonstrated by the following figure.

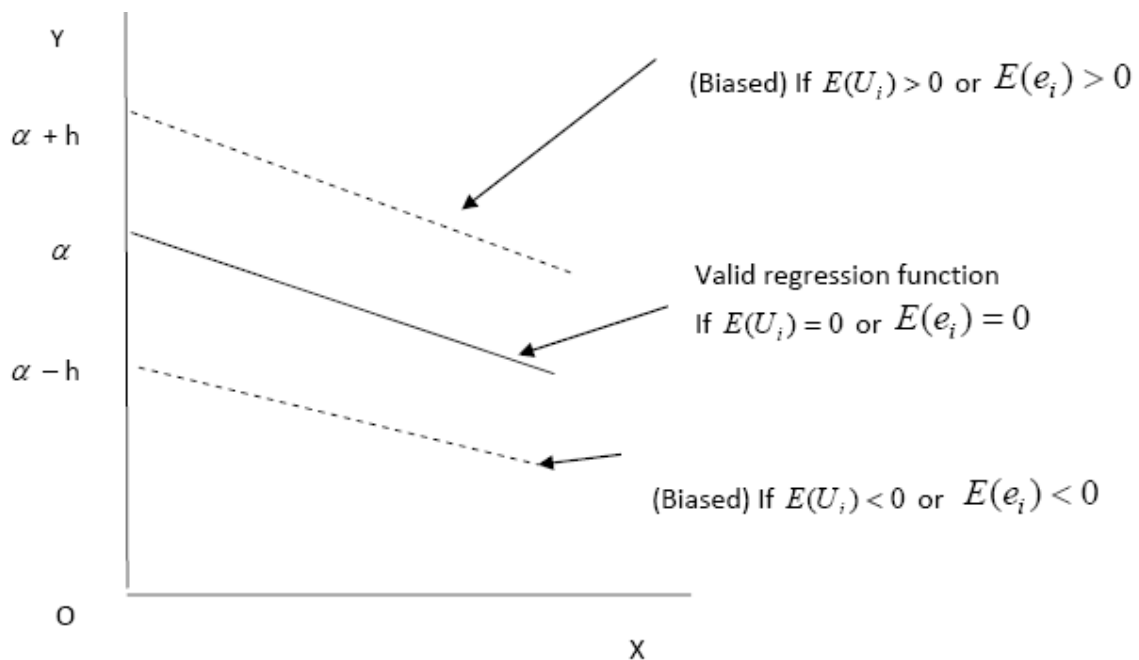


Figure 2: Regression Function/curve if the mean of error term is not zero

3. The disturbance terms have constant variance in each period. This is given as follows:

$$Var(U_i) = E(U_i - E(U_i))^2 = \sigma_u^2$$

. This assumption is known as the assumption of homoscedasticity. If this condition is not fulfilled or if the variance of the error terms varies as sample size changes or as the value of explanatory variables changes, then this leads to Heteroscedasticity problem.

4. Explanatory variables  $X_i$  and disturbance terms  $U_i$  are uncorrelated or independent. All the co-variances of the successive values of the error term are equal to zero. This condition is given by  $Cov(U_i, X_i) = 0$ . It is followed from this that the following identity holds true;  $\sum e_i X_i = 0$ . The value in which the error term assumed in one period does not depend on the value in which it assumed in any other period. If this condition is not met by our data or variables, our regression function and conclusions to be drawn from it will be invalid. This assumption is known as the assumption of non-autocorrelation or non-serial correlation.
5. The explanatory variable  $X_i$  is fixed in repeated samples. Each value of  $X_i$  does not vary for instance owing to change in sample size. This means the explanatory variables are non-random and hence distributional free variable.
6. Linearity of the model in parameters. The simple linear regression requires linearity in parameters; but not necessarily linearity in variables. The same technique can be applied to estimate regression functions of the following forms:  $Y = f(X)$ ;  $Y = f(X^2)$ ;  $Y = f(X^3)$ ;  $Y = f(X - kX)$ ; and so on. What is important is transforming the data as required.
7. Normality assumption-The disturbance term  $U_i$  is assumed to have a normal distribution with zero mean and a constant variance. This assumption is given as follows:

$$U_i \sim N(0, \sigma_u^2)$$

. This assumption is a combination of zero mean of error term assumption and homoscedasticity assumption. This assumption or combination of assumptions is used in testing hypotheses about significance of parameters. It is also useful in both estimating parameters and testing their significance in maximum likelihood method.

8. Explanatory variables should not be perfectly, linearly and/or highly correlated. Using explanatory variables which are highly or perfectly correlated in a regression function causes a biased function or model. It also results in multicollinearity problem.
9. The relationship between variables (or the model) is correctly specified. For instance all the necessary variables are included in the model. The variables are in the form that best describes the functional relationship. For instance,  $Y = f(X^2)$  may better reflect the relationship between Y and X than  $Y = f(X)$ .

10. The explanatory variables do not have identical value. This assumption is very important for improving the precision of estimators.

Note that some of these assumptions or conditions, (those which imply to more than one explanatory variables), are meant for the next chapters (along with all the other assumptions or conditions). So, we may not restate these conditions in the next chapter even if they are required there also.

## ii) OLS method of Estimation

Estimating a linear regression function using the Ordinary Least Square (OLS) method is simply about calculating the parameters of the regression function for which the sum of square of the error terms is minimized. The procedure is given as follows. Suppose we want to estimate the following equation

$$Y_i = \beta_0 + \beta_1 X_i + U_i$$

Since most of the time we use sample (or it is difficult to get population data) the corresponding sample regression function is given as follows.

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$$

From this identity, we solve for the residual term

$$e_i = Y_i - \hat{Y}_i$$

, square both sides and then take

sum of both sides. These three steps are given (respectively as follows.

$$e_i = Y_i - \hat{Y}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i \quad 2.7$$

$$\sum e_i^2 = \sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2 \quad 2.8$$

Where,

$$\sum e_i^2 = \text{RSS}$$

Residual Sum of Squares.

The method of OLS involves finding the estimates of the intercept and the slope for which the sum squares given by the Equation is minimized. To minimize the residual sum of squares we take the first order partial derivatives of Equation 2.8 and equate them to zero.

That is, the partial derivative with respect to  $\hat{\beta}_0$  :

$$\frac{\partial \sum e_i^2}{\partial \hat{\beta}_0} = 2 \sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)(-1) = 0 \quad 2.9$$

$$\Rightarrow \sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0$$

2.10

$$\Rightarrow \sum Y_i - n\hat{\beta}_0 - \hat{\beta}_1 \sum X_i = 0 \quad 2.11$$

$$\Rightarrow \sum Y_i = n\hat{\beta}_0 + \hat{\beta}_1 \sum X_i \quad 2.12$$

Where n is the sample size.

Partial derivative With respect to



$$\frac{\partial}{\partial \hat{\beta}_1}$$

2.13

$$\Rightarrow \sum (Y_i X_i - \hat{\beta}_0 X_i - \hat{\beta}_1 X_i^2) = 0$$

2.14

$$\Rightarrow \sum X_i Y_i - \hat{\beta}_0 \sum X_i - \hat{\beta}_1 \sum X_i^2 = 0 \quad 2.15$$

Note that the equation  $\sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2$  is a composite function and we should apply a chain rule in finding the partial derivatives with respect to the parameter estimates.

Equations 2.12 and 2.15 are together called the system of normal equations. Solving the system of normal equations simultaneously we obtain:

$$\hat{\beta}_1 = \frac{n \sum XY - (\sum X)(\sum Y)}{n \sum X^2 - (\sum X)^2} \quad \text{Or}$$

$$\hat{\beta}_1 = \frac{\sum XY - n \bar{Y} \bar{X}}{\sum X_i^2 - n \bar{X}^2} \quad \text{and we have } \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} \quad \text{from above}$$

Now we have the formula to estimate the simple linear regression function. Let us illustrate with example.

Example 2.4: Given the following sample data of three pairs of Y (dependent variable) and X (independent variable), find a simple linear regression function;  $Y = f(X)$ .

| $Y_i$ | $X_i$ |
|-------|-------|
| 10    | 30    |
| 20    | 50    |
| 30    | 60    |

- find a simple linear regression function;  $Y = f(X)$
- Interpret your result.

c) Predict the value of Y when X is 45.

**Solution:**

a. To fit the regression equation we do the following computations.

|      | $Y_i$                               | $X_i$ | $Y_i X_i$ | $X_i^2$ |
|------|-------------------------------------|-------|-----------|---------|
|      | 10                                  | 30    | 300       | 900     |
|      | 20                                  | 50    | 1000      | 2500    |
|      | 30                                  | 60    | 1800      | 3600    |
| Sum  | 60                                  | 140   | 3100      | 7000    |
| Mean | $\bar{X} = \frac{140}{3}$<br>$= 20$ |       |           |         |

$$\hat{\beta}_1 = \frac{n \sum XY - (\sum X)(\sum Y)}{n \sum X^2 - (\sum X)^2} = \frac{3(3100) - (140)(60)}{3(7000) - (140)^2} = 0.64$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} = 20 - 0.64(140/3) = -10$$

Thus the fitted regression function is given by:  $\hat{Y}_i = -10 + 0.64 X_i$

b) Interpretation, the value of the intercept term, -10, implies that the value of the dependent variable Y is -10 when the value of the explanatory variable is zero. The value of the slope coefficient ( $\hat{\beta}_1 = 0.64$ ) is a measure of the marginal change in the dependent variable Y when the value of the explanatory variable increases by one. For instance, in this model, the value of Y increases on average by 0.64 units when X increases by one.

c)  $\hat{Y}_i = -10 + (0.64)(45) = 18.8$

That means when X assumes a value of 45, the value of Y on average is expected to be 18.8. The regression coefficients can also be obtained by simple formulae by taking the deviations between the original values and their means. Now, if

$$x_i = X_i - \bar{X}, \text{ and } y_i = Y_i - \bar{Y}$$

Then, the coefficients can be represented by alternative formula given below  $\hat{\beta}_1 = \frac{\sum x_i y_i}{\sum x_i^2}$ , and  $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$

Example 2.5: Find the regression equation for the data under Example 2.4, using the shortcut formula. To solve this problem we proceed as follows.

|     | $Y_i$ | $X_i$ | y   | x      | xy     | $x^2$  | $y^2$ |
|-----|-------|-------|-----|--------|--------|--------|-------|
|     | 10    | 30    | -10 | -16.67 | 166.67 | 277.78 | 100   |
|     | 20    | 50    | 0   | 3.33   | 0.00   | 11.11  | 0     |
|     | 30    | 60    | 10  | 13.33  | 133.33 | 177.78 | 100   |
| Sum | 60    | 140   | 0   | 0      | 300.00 | 466.67 | 200   |

|      |    |          |  |  |  |  |  |
|------|----|----------|--|--|--|--|--|
| Mean | 20 | 46.66667 |  |  |  |  |  |
|------|----|----------|--|--|--|--|--|

Then

$$\hat{\beta}_1 = \frac{\sum x_i y_i}{\sum x_i^2} = \frac{300}{466.67} = 0.64, \text{ and } \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} = 20 - (0.64)(46.67) = -10 \text{ with results similar to previous case.}$$

## Mean and Variance of Parameter Estimates

We have seen that how the numerical values of the parameter estimates can be obtained using OLS estimating techniques. Now let us see their distributional nature, i.e. the mean and variance of the parameter estimates. There can be several samples of the same size that can be drawn from the same population. For each sample, the parameter estimates have their own specific numerical values. That means the values of the estimates are different when we go from one sample to another. Therefore, the parameter estimate have different values for estimating a given true population parameter. That means the parameter estimates are random in their nature and should have distinct distribution with the corresponding parameters. Remember we have discussed in the previous sections that both the error term and the dependent variable are assumed to be normally distributed. Thus, the parameter estimates also have a normal distribution with their associative mean and variance. Formula for mean and variance of the respective parameter estimates and the error term are given below (procedure to derive is given in Annex A)

1. The mean of  $\hat{\beta}_1 = E(\hat{\beta}_1) = \beta_1$
2. The variance of  $\hat{\beta}_1 = Var(\hat{\beta}_1) = E((\hat{\beta}_1 - E(\hat{\beta}_1))^2) = \frac{\sigma_u^2}{\sum x_i^2}$
3. The mean of  $\hat{\beta}_0 = E(\hat{\beta}_0) = \beta_0$
4. The variance of  $\hat{\beta}_0 = \frac{\sigma_u^2}{\sum x_i^2}$
5. The estimated value of the variance of the error term

$$\hat{\sigma}_u^2 = \frac{\sum e_i^2}{n - 2}$$

## Hypothesis Testing of OLS Estimates

After estimation of the parameters there are important issues to be considered by the researcher. We have to know that to what extent our estimates are reliable enough and acceptable for further purpose. That means, we have to evaluate the degree of representativeness of the estimate to the true population parameter. Simply a model must be tested for its significance before it can be used for any other purpose. In this subsection we will evaluate the reliability of model estimated using the procedure we explained above.

The available test criteria are divided in to three groups: Theoretical a priori criteria, statistical criteria and econometric criteria. Priori criteria set by economic theories are in line with the consistency of coefficients of econometric model to the economic theory. Statistical criteria,

also known as first order tests, are set by statistical theory and refer to evaluate the statistical reliability of the model. Econometric criteria refer to whether the assumptions of an econometric model employed in estimating the parameters are fulfilled or not. There are two most commonly used tests in econometrics. These are:

1. The square of correlation coefficient (  $r^2$  ) which is used for judging the explanatory power of the linear regression of Y on X or on X's. The square of correlation coefficient in simple regression is known as the coefficient of determination and is given by  $R^2$ . The coefficient of determination measures the goodness of fit of the line of regression on the observed sample values of Y and X.
2. The standard error test of the parameter estimates applied for judging the statistical reliability of the estimates. This test measures the degree of confidence that we may attribute to the estimates.

### i) The Coefficient of determination ( $R^2$ )

The coefficient of determination is the measure of the amount or proportion of the total variation of the dependent variable that is determined or explained by the model or the presence of the explanatory variable in the model. The total variation of the dependent variable is split in two additive components; a part explained by the model and a part represented by the random term. The total variation of the dependent variable is measured from its arithmetic mean.

$$\begin{aligned}\text{Total variation} &= T = \sum (Y - \bar{Y})^2 \\ \text{Total explained variation} &= \sum (\hat{Y} - \bar{Y})^2 \\ \text{Total unexplained variation} &= \sum e_i^2\end{aligned}$$

The total variation of the dependent variable is given in the following form; TSS=ESS + RSS, which means total sum of square of the dependent variable is split into explained sum of square and residual sum of square.

$$\begin{aligned}e_i &= y_i - \hat{y}_i \\ y_i &= \hat{y}_i + e_i \\ y_i^2 &= \hat{y}_i^2 + e_i^2 + 2\hat{y}_i e_i \\ \sum y_i^2 &= \sum \hat{y}_i^2 + \sum e_i^2 + 2\sum \hat{y}_i e_i \\ \text{But } \sum \hat{y}_i e_i &= 0 \\ \text{Therefore, } \sum y_i^2 &= \sum \hat{y}_i^2 + \sum e_i^2\end{aligned}$$

The coefficient of determination is given by the formula

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$$R^2 = \frac{\sum (\hat{Y}_i - \bar{Y})^2}{\sum (Y_i - \bar{Y})^2} = \frac{\sum \hat{y}_i^2}{\sum y_i^2} = \frac{\text{Explained Variation in Y}}{\text{Total Variation in Y}} \quad 2.16$$

Since  $\sum \hat{y}_i^2 = \hat{\beta}_1 \sum x_i y_i$   
 $R^2 = \frac{\hat{\beta}_1 \sum x_i y_i}{\sum y_i^2}$  the coefficient of determination can also be given as

Or

$$R^2 = 1 - \frac{\text{Unexplained Variation in Y}}{\text{Total Variation in Y}} = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum y_i^2} = \frac{\sum \hat{y}_i^2}{\sum y_i^2} \quad 2.17$$

The higher the coefficient of determination is the better the fit. Conversely, the smaller the coefficient of determination is the poorer the fit. That is why the coefficient of determination is used to compare two or more models. One minus the coefficient of determination is called the coefficient of non-determination, and it gives the proportion of the variation in the dependent variable that remained undetermined or unexplained by the model.

## ii) Testing the significance of a given regression coefficient

Since the sample values of the intercept and the coefficient are estimates of the true population parameters, we have to test them for their statistical reliability.

The significance of a model can be seen in terms of the amount of variation in the dependent variable that it explains and the significance of the regression coefficients.

There are different tests that are available to test the statistical reliability of the parameter estimates. The following are the common ones;

- A) The standard error test
- B) The standard normal test
- C) The students t-test

Now, let us discuss them one by one.

### A) The Standard Error Test

This test first establishes the two hypotheses that are going to be tested which are commonly known as the null and alternative hypotheses. The null hypothesis addresses that the sample is coming from the population whose parameter is not significantly different from zero while the alternative hypothesis addresses that the sample is coming from the population whose parameter is significantly different from zero. The two hypotheses are given as follows:

$$H_0: \beta_i = 0$$

$$H_1: \beta_i \neq 0$$

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The standard error test is outlined as follows:

1. Compute the standard deviations of the parameter estimates using the above formula for variances of parameter estimates. This is because standard deviation is the positive square root of the variance.

$$se(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{\sum x_i^2}}$$

$$se(\hat{\beta}_2) = \sqrt{\frac{\hat{\sigma}^2}{\sum y_i^2}}$$

2. Compare the standard errors of the estimates with the numerical values of the estimates and make decision.

A) If the standard error of the estimate is less than half of the numerical value of the estimate, we can conclude that the estimate is statistically significant. That is, if  $se(\hat{\beta}_i) < \frac{1}{2}(\hat{\beta}_i)$ , reject the null hypothesis and we can conclude that the estimate is statistically significant.

B) If the standard error of the estimate is greater than half of the numerical value of the estimate, the parameter estimate is not statistically reliable. That is, if  $se(\hat{\beta}_i) > \frac{1}{2}(\hat{\beta}_i)$ , conclude to accept the null hypothesis and conclude that the estimate is not statistically significant.

### B) The Standard Normal Test

This test is based on the normal distribution. The test is applicable if:

- The standard deviation of the population is known irrespective of the sample size
- The standard deviation of the population is unknown provided that the sample size is sufficiently large ( $n > 30$ ).

The standard normal test or Z-test is outline as follows;

1. Test the null hypothesis  $H_0 : \beta_i = 0$  against the alternative hypothesis  $H_1 : \beta_i \neq 0$

2. Determine the level of significant (  $\alpha$  ) in which the test is carried

out. It is the probability of committing type I error, i.e. the probability of rejecting the null hypothesis while it is true. It is common in applied econometrics to use 5% level of significance.

3. Determine the theoretical or tabulated value of Z from the table. That is, find the value of  $Z_{\alpha/2}$  from the standard normal table.  $Z_{0.025} = 1.96$  from the table.

4. Make decision. The decision of statistical hypothesis testing consists of two decisions; either accepting the null hypothesis or rejecting it.

If  $|Z_{cal}| < Z_{\alpha/2}$ , accept the null hypothesis while if  $|Z_{cal}| > Z_{\alpha/2}$ , reject the null hypothesis. It is true that most of the times the null and alternative hypotheses are mutually exclusive. Accepting the null hypothesis means that rejecting

the alternative hypothesis and rejecting the null hypothesis means accepting the alternative hypothesis.

**Example:** If the regression has a value of  $\hat{\beta}_1 = 29.48$  and the standard error of  $\hat{\beta}_1$  is 36. Test the hypothesis that the value of  $\beta_1 = 25$  at 5% level of significance using standard normal test.

**Solution:** We have to follow the procedures of the test.

$$H_0 : \beta_1 = 25$$

$$H_1 : \beta_1 \neq 25$$

After setting up the hypotheses to be tested, the next step is to determine the level of significance in which the test is carried out. In the above example the significance level is given as 5%.

The third step is to find the theoretical value of Z at specified level of significance. From the standard normal table we can get that  $Z_{0.025} = 1.96$ .

The fourth step in hypothesis testing is computing the observed or calculated value of the standard normal distribution using the following formula.

$Z_{cal} = \frac{\hat{\beta}_1 - \beta_1}{se(\hat{\beta}_1)} = \frac{29.48 - 25}{36} = 0.12$ . Since the calculated value of the test statistic is less than the tabulated value, the decision is to accept the null hypothesis and conclude that the value of the parameter is 25.

### C) The Student t-Test

In conditions where Z-test is not applied (in small samples), t-test can be used to test the statistical reliability of the parameter estimates. The test depends on the degrees of freedom that the sample has. The test procedures of t-test are similar with that of the z-test. The procedures are outlined as follows;

1. Set up the hypothesis. The hypotheses for testing a given regression coefficient is given by:

$$H_0 : \beta_i = 0$$

$$H_1 : \beta_i \neq 0$$

2. Determine the level of significance for carrying out the test. We usually use a 5% level significance in applied econometric research.
3. Determine the tabulated value of t from the table with n-k degrees of freedom, where k is the number of parameters estimated.
4. Determine the calculated value of t. The test statistic (using the t-test) is given by:

$$t_{cal} = \frac{\hat{\beta}_i}{se(\hat{\beta}_i)}$$

The test rule or decision is given as follows:

Reject  $H_0$  if  $|t_{cal}| \geq t_{\alpha/2, n-k}$

### iii) Confidence Interval Estimation of the regression Coefficients

We have discussed the important tests that can be conducted to check model and parameters validity. But one thing that must be clear is that rejecting the null hypothesis does not mean that the parameter estimates are correct estimates of the true population parameters. It means that the estimate comes from the sample drawn from the population whose population parameter is significantly different from zero. In order to define the range within which the true parameter lies, we must construct a confidence interval for the parameter. Like we constructed confidence interval estimates for a given population mean, using the sample mean (in Introduction to Statistics), we can construct 100(1-  $\alpha$ ) % confidence intervals for the sample regression coefficients. To do so we need to have the standard errors of the sample regression coefficients. The standard error of a given coefficient is the positive square root of the variance of the coefficient. Thus, we have discussed that the formulae for finding the variances of the regression coefficients are given as.

$$\text{Variance of the intercept is given by } \text{var}(\hat{\beta}_0) = \frac{\sum X_i^2}{n \sum x_i^2} \quad 2.18$$

$$\text{Variance of the slope } \beta_1 \text{ is given by } \frac{\sum y_i^2}{\sum x_i^2} \quad 2.19$$

Where,  $\frac{\sum y_i^2}{\sum x_i^2}$  (2.20) is the estimate of the variance of the random term and k is the number of parameters to be estimated in the model. The standard errors are the positive square root of the variances and the 100 (1-  $\alpha$ ) % confidence interval for the slope is given by:

$$\hat{\beta}_1 - t_{\alpha/2, n-k} (se(\hat{\beta}_1)) \leq \beta_1 \leq \hat{\beta}_1 + t_{\alpha/2, n-k} (se(\hat{\beta}_1))$$

$$\beta_1 = \hat{\beta}_1 \pm t_{\alpha/2, n-k} (se(\hat{\beta}_1)) \quad 2.21$$

And for the intercept:

$$\hat{\beta}_0 = \hat{\beta}_0 \pm t_{\alpha/2, n-k} (se(\hat{\beta}_0)) \quad 2.22$$

Example 2.6: The following table gives the quantity supplied (Y in tons) and its price (X pound per ton) for a commodity over a period of twelve years.

Table 5: Data on supply and price for given commodity

|   |    |    |    |    |    |    |    |    |    |    |    |    |
|---|----|----|----|----|----|----|----|----|----|----|----|----|
| Y | 69 | 76 | 52 | 56 | 57 | 77 | 58 | 55 | 67 | 53 | 72 | 64 |
| X | 9  | 12 | 6  | 10 | 9  | 10 | 7  | 8  | 12 | 6  | 11 | 8  |

Table 6: Data for computation of different parameters

| Time | Y   | X   | XY   | X <sup>2</sup> | Y <sup>2</sup> | x  | Y  | Xy  | x <sup>2</sup> | y <sup>2</sup> | $\hat{Y}$ | e <sub>i</sub> | e <sub>i</sub> <sup>2</sup> |
|------|-----|-----|------|----------------|----------------|----|----|-----|----------------|----------------|-----------|----------------|-----------------------------|
| 1    | 69  | 9   | 621  | 81             | 4761           | 0  | 6  | 0   | 0              | 36             | 63.00     | 6.00           | 36.00                       |
| 2    | 76  | 12  | 912  | 144            | 5776           | 3  | 13 | 39  | 9              | 16             | 72.75     | 3.25           | 10.56                       |
| 3    | 52  | 6   | 312  | 36             | 2704           | -3 | -1 | 33  | 9              | 12             | 53.25     | -1.25          | 1.56                        |
| 4    | 56  | 10  | 560  | 100            | 3136           | 1  | -7 | -7  | 1              | 49             | 66.25     | -10.25         | 105.06                      |
| 5    | 57  | 9   | 513  | 81             | 3249           | 0  | -6 | 0   | 0              | 36             | 63.00     | -6.00          | 36.00                       |
| 6    | 77  | 10  | 770  | 100            | 5929           | 1  | 14 | 14  | 1              | 19             | 66.25     | 10.75          | 115.56                      |
| 7    | 58  | 7   | 406  | 49             | 3364           | -2 | -5 | 10  | 4              | 25             | 56.50     | 1.50           | 2.25                        |
| 8    | 55  | 8   | 440  | 64             | 3025           | -1 | -8 | 8   | 1              | 64             | 59.75     | -4.75          | 22.56                       |
| 9    | 67  | 12  | 804  | 144            | 4489           | 3  | 4  | 12  | 9              | 16             | 72.75     | -5.75          | 33.06                       |
| 10   | 53  | 6   | 318  | 36             | 2809           | -3 | -1 | 30  | 9              | 10             | 53.25     | -0.25          | 0.06                        |
| 11   | 72  | 11  | 792  | 121            | 5184           | 2  | 9  | 18  | 4              | 81             | 69.50     | 2.50           | 6.25                        |
| 12   | 64  | 8   | 512  | 64             | 4096           | -1 | 1  | -1  | 1              | 1              | 59.75     | 4.25           | 18.06                       |
| Sum  | 756 | 108 | 6960 | 1020           | 48520          | 0  | 0  | 150 | 40             | 890            | 756.0     | 0.00           | 387.0                       |

Use Tables (Table 5 and Table 6) to answer the following questions

1. Estimate the Coefficient of determination ( $R^2$ )
2. Run significance test of regression coefficients using the following test methods
  - A) The standard error test
  - B) The students t-test
3. Fit the linear regression equation and determine the 95% confidence interval for the slope.

### Solution

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1. Estimate the Coefficient of determination ( $R^2$ )

Refer to Example 2.6 above to determine how much percent of the variations in the quantity supplied is explained by the price of the commodity and what percent remained unexplained.

Use data in Table 6 to estimate  $R^2$  using the formula given below.

$$R^2 = 1 - \frac{\sum e_i^2}{\sum y_i^2} = 1 - \frac{387}{894} = 1 - 0.43 = 0.57$$

This result shows that 57% of the variation in the quantity supplied of the commodity under consideration is explained by the variation in the price of the commodity; and the rest 37% remain unexplained by the price of the commodity. In other word, there may be other important explanatory variables left out that could contribute to the variation in the quantity supplied of the commodity, under consideration.

2. Run significance test of regression coefficients using the following test methods

Fitted regression line for the data given is:

$$\hat{Y}_i = 33.75 + 3.25X_i, \text{ where the numbers in parenthesis } \begin{pmatrix} 8.3 \\ 0.9 \end{pmatrix} \text{ are standard errors of the respective coefficients.}$$

#### A. Standard Error test

In testing the statistical significance of the estimates using standard error test, the following information needed for decision.

Since there are two parameter estimates in the model, we have to test them separately.

Testing for  $\hat{\beta}_1$

We have the following information about  $\hat{\beta}_1$  i.e  $\hat{\beta}_1 = 3.25$  and  $se(\hat{\beta}_1) = 0.9$

The following are the null and alternative hypotheses to be tested.

$$\begin{aligned} H_0 : \beta_1 &= 0 \\ H_1 : \beta_1 &\neq 0 \end{aligned}$$

Since the standard error of  $\hat{\beta}_1$  is less than half of the value of  $\hat{\beta}_1$ , we have to reject the null hypothesis and conclude that the parameter estimate  $\hat{\beta}_1$  is statistically significant.

Testing for  $\hat{\beta}_0$

Again we have the following information about  $\hat{\beta}_0$

$$\hat{\beta}_0 = 33.75 \text{ and } se(\hat{\beta}_0) = 8.3$$

The hypotheses to be tested are given as follows;

$$H_0 : \beta_0 = 0$$

$$H_1 : \beta_0 \neq 0$$

Since the standard error of  $\hat{\beta}_0$  is less than half of the numerical value of  $\hat{\beta}_0$ , we have to reject the null hypothesis and conclude that  $\hat{\beta}_0$  is statistically significant.

#### B. The students t-test

In the illustrative example, we can apply t-test to see whether price of the commodity is significant in determining the quantity supplied of the commodity under consideration? Use  $\alpha=0.05$ .

The hypothesis to be tested is:

$$H_0: \beta = 0$$

$$H_1: \beta \neq 0$$

The parameters are known.

Then we can estimate  $t_{cal}$  as follows

$$t_{cal} = \frac{\hat{\beta}_0}{SE(\hat{\beta}_0)}$$

Further tabulated value for t is 2.228. When we compare these two values, the calculated t is greater than the tabulated value. Hence, we reject the null hypothesis. Rejecting the null hypothesis means, concluding that the price of the commodity is significant in determining the quantity supplied for the commodity.

In this part we have seen how to conduct the statistical reliability test using t-statistic. Now let us see additional information about this test. When the degrees of freedom is large, we can conduct t-test without consulting the t-table in finding the theoretical value of t. This rule is known as 2t-rule. The rule is stated as follows;

The t-table shows that the values of t changes very slowly if the degrees of freedom (n-k) are greater than 8. For example the value of  $t_{0.025}$  changes from 2.30 (when n-k=8) to 1.96 (when n-k=∞). The change from 2.30 to 1.96 is obviously very slow. Consequently, we can ignore the degrees of freedom (when they are greater than 8) and say that the theoretical value of  $t_{cal}$  is 2.0. Thus, a two tail test of a null hypothesis at 5% level of significance can be reduced to the following rules.

1. If  $t_{cal}$  is greater than 2 or less than -2, we reject the null hypothesis
2. If  $t_{cal}$  is less than 2 or greater than -2, accept the null hypothesis.
3. Fit the linear regression equation and determine the 95% confidence interval for the slope.

Fitted regression model is indicated  $\hat{y} = 1.23 + 0.05x$ , where the numbers in parenthesis are standard errors of the respective coefficients. To estimate confidence interval we need standard error which is determined as follows

$$SE(\hat{\beta}_1) = \frac{SE(y)}{\sqrt{\sum x^2}}$$

$$var(\hat{\beta}_1) = \sigma^2 \frac{1}{\sum x^2} = 38.7 \left( \frac{1}{48} \right) = 0.80625$$

The standard error of the slope is  $SE(\hat{\beta}_1) = 0.898$

The tabulated value of  $t$  for degrees of freedom  $12-2=10$  and  $\alpha/2=0.025$  is 2.228.

Hence the 95% confidence interval for the slope is given by:

 The result tells us that at the error probability 0.05, the true value of the slope coefficient lies between 1.25 and 5.25

### **Properties of OLS Estimators**

The ideal or optimum properties that the OLS estimates possess may be summarized by well known theorem known as the Gauss-Markov Theorem.

Statement of the theorem: Given the assumptions of the classical linear regression model, the OLS estimators, in the class of linear and unbiased estimators, have the minimum variance, i.e. the OLS estimators are BLUE.

According to this theorem, under the basic assumptions of the classical linear regression model, the least squares estimators are linear, unbiased and have minimum variance (i.e. are best of all linear unbiased estimators). Sometimes the theorem is referred to as the BLUE theorem i.e. Best, Linear, Unbiased Estimator. An estimator is called BLUE if:

- a. Linear: a linear function of the random variable, such as, the dependent variable  $Y$ .
- b. Unbiased: its average or expected value is equal to the true population parameter.
- c. Minimum variance: It has a minimum variance in the class of linear and unbiased estimators. An unbiased estimator with the least variance is known as an efficient estimator.

### **Statistical Software Application Practices**

## CHAPTER FOUR

### MULTIPLE REGRESSION ANALYSIS

#### Assumptions of the Multiple Linear Regression

Each econometric method that would be used for estimation purpose has its own assumptions. Knowing the assumptions and their consequence if they are not maintained is very important for the econometrician. In the previous section, there are certain assumptions underlying the multiple regression model, under the method of ordinary least squares (OLS). Let us see them one by one.

**Assumption 1:** Randomness of  $u_i$  - the variable  $u$  is a real random variable.

**Assumption 2:** Zero mean of  $u_i$  - the random variable  $u_i$  has a zero

mean for each value of  $X_i$  i.e.  $E(u_i) = 0$

**Assumption 3:** Homoscedasticity of the random term - the random term  $u_i$  has constant variance. In other words, the variance of each  $u_i$  is the same for all the  $X_i$  values.

$$E(u_i^2) = \sigma_u^2 \text{ Constant}$$

**Assumption 4:** Normality of  $u_i$  - the values of each  $u_i$  are normally distributed  $u_i \sim N(0, \sigma_u^2)$

**Assumption 5:** No autocorrelation or serial independence of the random terms - the successive values of the random term are not strongly correlated. The values of

$u_i$  (corresponding to  $x_i$ ) are independent of the values of any other  $u_j$  (corresponding to

$x_j$ ).

**Assumption 6:** Independence of  $u_i$  and  $X_i$  - every disturbance term  $u_i$  is independent of the explanatory variables.  $E(u_i | X_i) = 0$

**Assumption 7:** No errors of measurement in the explanatory variables are measured without error.

**Assumption 8:** No perfect multicollinearity among the  $X_i$ 's - the explanatory variables are not perfectly linearly correlated.

**Assumption 9:** Correct specification of the model - the model has no specification error in that all the important explanatory variables appear explicitly in the function and the mathematical form is correctly defined (linear or non-linear form and the number of equations in the model).

### Estimation of Partial Regression Coefficients

The process of estimating the parameters in the multiple regression model is similar with that of the simple linear regression model. The main task is to derive the normal equations using the same procedure as the case of simple regression. Like in the simple linear regression model case, OLS and Maximum Likelihood (ML) methods can be used to estimate partial regression coefficients of multiple regression models. But, due to their simplicity and popularity, OLS methods can be used. The OLS procedure consists in so choosing the values of the unknown parameters that the residual sum of squares is as small as possible.

Under the assumption of zero mean of the random term, the sample regression function will look like the following.

$$\hat{Y}_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_k X_{ki} \quad (3.3)$$

We call this equation, the fitted equation. Subtracting (3.3) from (3.2), we obtain:

$$e_i = Y_i - \hat{Y}_i \quad (3.4)$$

The method of ordinary least squares (OLS) or classical least square (CLS) involves obtaining the values of  $\beta_0, \beta_1, \beta_2, \dots, \beta_k$  is minimum.

The values of  $\sum e_i^2$  is minimum is obtained by differentiation this sum of squares with respect to these coefficients and equate them to zero. That is,

$$\frac{\partial \sum e_i^2}{\partial \beta_0} = 0$$

$$\frac{\partial \sum e_i^2}{\partial \beta_1} = 0$$

$$\frac{\partial \sum e_i^2}{\partial \beta_2} = 0$$

Solving equations (3.5), (3.6) and (3.7) simultaneously, we obtain the system of normal equations given as follows:

$$\sum Y_i = n\beta_0 + \beta_1 \sum X_{1i} + \beta_2 \sum X_{2i} + \dots + \beta_k \sum X_{ki}$$

Then, letting

$$\sum Y_i = n\beta_0 + \beta_1 \sum X_{1i} + \beta_2 \sum X_{2i} + \dots + \beta_k \sum X_{ki}$$

$$\begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} = \begin{pmatrix} \sum y_i \\ \sum x_i y_i \end{pmatrix} \begin{pmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{pmatrix}^{-1}$$

The above three equations (3.8), (3.9) and (3.10) can be solved using Matrix operations or simultaneously to obtain the following estimates:

$$\hat{\beta}_0 = \frac{(\sum x_i^2)(\sum y_i) - (\sum x_i)(\sum x_i y_i)}{(\sum x_i^2)(\sum y_i) - (\sum x_i)(\sum x_i y_i)} \quad 3.14$$

$$\hat{\beta}_1 = \frac{(\sum x_i y_i)(\sum x_i^2) - (\sum x_i)(\sum x_i y_i)}{(\sum x_i^2)(\sum y_i) - (\sum x_i)(\sum x_i y_i)} \quad 3.15$$

$$\hat{\beta}_1 = \bar{y} - \hat{\beta}_0 \bar{x}_1 - \hat{\beta}_2 \bar{x}_2 \quad 3.16$$

## Variance and Standard errors of OLS Estimators

Estimating the numerical values of the parameters is not enough in econometrics if the data are coming from the samples. The standard errors derived are important for two main purposes: to establish confidence intervals for the parameters and to test statistical hypotheses. They are important to look into their precision or statistical reliability. An estimator cannot be used for any purpose if it is not a good estimator. The precision of an estimator is measured by observing the standard error of the estimator.

Like in the case of simple linear regression, the standard errors of the coefficients are vital in statistical inferences about the coefficients. We use standard the error of a coefficient to construct confidence interval estimate for the population regression coefficient and to test the significance of the variable to which the coefficient is attached in determining the dependent variable in the model. In this section, we will see these standard errors. The standard error of a coefficient is the positive square root of the variance of the coefficient. Thus, we start with defining the variances of the coefficients.

Variance of the intercept  $\left( \hat{\beta}_0 \right)$

$$\text{Var}(\hat{\beta}_0) = \sigma^2 \left[ \frac{1}{n} + \frac{\bar{x}^2 \sum x_i^2 + \bar{x}^2 \sum x_i^2 - 2\bar{x} \sum x_i x_i}{\sum x_i^2 \sum x_i^2 - (\sum x_i x_i)^2} \right] \quad 3.17$$

Variance of  $\hat{\beta}_1$

$$\text{Var}(\hat{\beta}_1) = \sigma^2 \left[ \frac{1}{\sum x_i^2} \right]$$

$$3.18$$

Variance of  $\hat{\beta}_2$

$$\hat{\sigma}_e^2 = \frac{\sum_{i=1}^n e_i^2}{n-3} \quad 3.19$$

Where,

$$\hat{\sigma}_e^2 = \frac{\sum_{i=1}^n e_i^2}{n-3} \quad 3.20$$

Equation 3.20 here gives the estimate of the variance of the random term. Then, the standard errors are computed as follows:

Standard error of  $\hat{\beta}_0$

$$SE(\hat{\beta}_0) = \sqrt{Var(\hat{\beta}_0)} \quad 3.21$$

Standard error of  $\hat{\beta}_1$

$$SE(\hat{\beta}_1) = \sqrt{Var(\hat{\beta}_1)} \quad 3.22$$

Standard error of  $\hat{\beta}_2$

$$SE(\hat{\beta}_2) = \sqrt{Var(\hat{\beta}_2)} \quad 3.23$$

**Note:** The OLS estimators of the multiple regression model have properties which are parallel to those of the two-variable model.

### Coefficient of Multiple Determination

In simple regression model we have discussed about the coefficient of determination and its interpretation. In this section, we will discuss the coefficient of multiple determination which has an equivalent role with that of the simple model. As coefficient of determination is the square of the simple correlation in simple model, coefficient of multiple determination is the square of multiple correlation coefficient.

The coefficient of multiple determination ( $R^2$ ) is the measure of the proportion of the variation in the dependent variable that is explained jointly by the independent variables in the model. One minus  $R^2$  is called the coefficient of non-determination. It gives the proportion of the variation in the dependent variable that remains unexplained by the independent variables in the model. As in the case of simple linear regression,  $R^2$  is the ratio of the explained variation to the total variation. Mathematically:

$$R^2 = \frac{\sum_{i=1}^n \hat{e}_i^2}{\sum_{i=1}^n e_i^2} \quad 35$$

Or  $R^2$  can also be given in terms of the slope coefficients

$\hat{\beta}_1$  and  $\hat{\beta}_2$  as

$$R^2 = \frac{\hat{\beta}_1 \sum x_1 y + \hat{\beta}_2 \sum x_2 y}{\sum y^2} \quad 3.25$$

In simple linear regression, the higher the  $R^2$  means the better the model is determined by the explanatory variable in the model. In multiple linear regression, however, every time we insert additional explanatory variable in the model, the  $R^2$  increases irrespective of the improvement in the goodness-of-fit of the model. That means high

$R^2$  may not imply that the model is good.

Thus, we adjust the  $R^2$  as follows:

$$R^2_{adj} = 1 - \frac{(1 - R^2)(n)}{(n - k - 1)} \quad 3.26$$

Where, k = the number of explanatory variables in the model.

In multiple linear regression, therefore, we better interpret the adjusted  $R^2$  than the ordinary or the unadjusted  $R^2$ . We have known that the value of  $R^2$  is always between zero and one. But the adjusted  $R^2$  can lie outside this range even to be negative.

In the case of simple linear regression,  $R^2$  is the square of linear correlation coefficient. Again as the correlation coefficient lies between -1 and +1, the coefficient of determination ( $R^2$ ) lies between 0 and 1. The  $R^2$  of multiple linear regression also lies between 0 and +1. The adjusted  $R^2$ , however, can sometimes be negative when the goodness of fit is poor. When the adjusted  $R^2$  value is negative, we considered it as zero and interpret as no variation of the dependent variable is explained by regressors.

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### Confidence Interval Estimation

Confidence interval estimation in multiple linear regression follows the same formulae and procedures that we followed in simple linear regression. You are, therefore, required to

practice finding the confidence interval estimates of the intercept and the slopes in multiple regression with two explanatory variables.

Please recall that  $100(1-$

$\alpha$  % confidence interval is given as

$$\hat{\beta}_i \pm t_{\alpha/2, n-k} se(\hat{\beta}_i) \quad \text{where } k \text{ is the number of parameters to be estimated or the}$$

number of variables (both dependent and explanatory)

*Interpretation of the confidence interval:* Values of the parameter lying in the interval are plausible with  $100(1-\alpha)\%$  confidence.

### Hypothesis Testing in Multiple Regression

Hypothesis testing is important to draw inferences about the estimates and to know how representative the estimates are to the true population parameter. Once we go beyond the simple world of the two-variable linear regression model, hypothesis testing assumes several interesting forms such as the following.

- a) Testing hypothesis about an individual partial regression coefficient;
- b) Testing the overall significance of the estimated multiple regression model (finding out if all the partial slope coefficients are simultaneously equal to zero);
- c) Testing if two or more coefficients are equal to one another;
- d) Testing that the partial regression coefficients satisfy certain restrictions
- e) Testing the stability of the estimated regression model over time or in different cross-sectional units
- f) Testing the functional form of regression models.

These and other types of hypotheses tests can be referred from different Econometrics books. For the case in point, we will confine ourselves to the major ones. <sup>37</sup>

#### Testing individual regression coefficients

The tests concerning the individual coefficients can be done using the standard error test or the t-test. In all the cases the hypothesis is stated as:

$$\begin{aligned} H_0 : \beta_1 &= 0 \\ H_1 : \beta_1 &\neq 0 \end{aligned}$$

$$\begin{aligned} H_0 : \hat{\beta}_2 &= 0 & H_1 : \hat{\beta}_2 &\neq 0 \\ H_0 : \hat{\beta}_2 &\neq 0 & H_1 : \hat{\beta}_2 &= 0 \end{aligned}$$

In a) we will like to test the hypothesis that  $X_1$  has no linear influence on  $Y$  holding other variables constant. In b) we test the hypothesis that  $X_2$  has no linear relationship with  $Y$  holding other factors constant. The above hypotheses will lead us to a two-tailed test however, one-tailed test might also be important. There are two methods for testing significance of individual regression coefficients.

**a) Standard Error Test:** Using the standard error test we can test the above hypothesis. Thus the decision rule is based on the relationship between the numerical value of the parameter and the standard error of the same.

(i) If  $S(\hat{\beta}_i) > \frac{1}{2} \hat{\beta}_i$ , we accept the null hypothesis, i.e. the estimate of  $\beta_i$  is not statistically significant.

**Conclusion:** The coefficient  $(\hat{\beta}_i)$  is not statistically significant. In other words, it does not have a significant influence on the dependent variable.

(ii) If  $S(\hat{\beta}_i) < \frac{1}{2} \hat{\beta}_i$ , we fail to accept  $H_0$ , i.e., we reject the null hypothesis in favour of the alternative hypothesis meaning the estimate of  $\beta_i$  has a significant influence on the dependent variable.

**Generalisation:** The smaller the standard error, the stronger is the evidence that the estimates are statistically significant.

### (b) t-test

The more appropriate and formal way to test the above hypothesis is to use the t-test. As usual we compute the t-ratios and compare them with the tabulated t-values and make our decision.

Therefore:

**Decision Rule:** accept  $H_0$  if

Otherwise, reject the null hypothesis. Rejecting  $H_0$  means, the coefficient being tested is significantly different from 0. Not rejecting  $H_0$ , on the other hand, means we don't have sufficient evidence to conclude that the coefficient is different from 0.

### Testing the Overall Significance of Regression Model

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Here, we are interested to test the overall significance of the observed or estimated regression line, that is, whether the dependent variable is linearly related to all of the explanatory variables. Hypotheses of such type are often called joint hypotheses. Testing the overall

significance of the model means testing the null hypothesis that none of the explanatory variables in the model significantly determine the changes in the dependent variable. Put in other words, it means testing the null hypothesis that none of the explanatory variables significantly explain the dependent variable in the model. This can be stated as:

$$H_0 : \beta_1 = \beta_2 = 0$$

$$H_1 : \beta_i \neq 0, \text{ at least for one } i.$$

The test statistic for this test is given by:

$$F_{cal} = \frac{\frac{\sum \hat{y}^2}{k-1}}{\frac{\sum e^2}{n-k}}$$

Where, k is the number of explanatory variables in the model.

The results of the overall significance test of a model are summarized in the analysis of variance (ANOVA) table as follows.

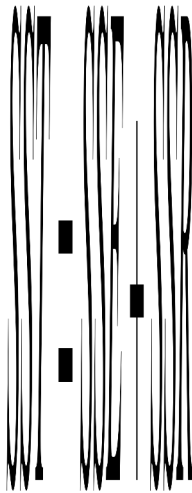
| Source of variation | Sum of squares             | Degrees of freedom | Mean sum of squares                    | $F_{cal}$               |
|---------------------|----------------------------|--------------------|----------------------------------------|-------------------------|
| Regression          | $\sum_{j=1}^k \hat{y}_j^2$ |                    | $\frac{\sum_{j=1}^k \hat{y}_j^2}{k-1}$ | $F = \frac{MSE}{MSE_R}$ |

|          |                            |       |                                              |  |
|----------|----------------------------|-------|----------------------------------------------|--|
| Residual | $SSA$                      | $n-k$ | $MSA = \frac{\sum_{i=1}^k \bar{y}_i^2}{n-k}$ |  |
| Total    | $SSY = \sum_{i=1}^n y_i^2$ | $n-1$ |                                              |  |

The values in this table are explained as follows:

$$\sum_{i=1}^n y_i^2 = \sum_{i=1}^n y_i^2$$

These three sums of squares are related in such a way that



This implies that the total sum of squares is the sum of the explained (regression) sum of squares and the residual (unexplained) sum of squares. In other words, the total variation in the dependent variable is the sum of the variation in the dependent variable due to the variation in the independent variables included in the model and the variation that remained unexplained by the explanatory variables in the model. Analysis of variance (ANOVA) is the technique of decomposing the total sum of squares into its components. As we can see here, the technique decomposes the total variation in the dependent variable into the explained and the unexplained variations. The degrees of freedom of the total variation are also the sum of the degrees of freedom of the two components. By dividing the sum of squares by the corresponding degrees of freedom, we obtain what is called the **Mean Sum of Squares (MSS)**.

The Mean Sum of Squares due to regression, errors (residual) and Total are calculated as the Sum of squares and the corresponding degrees of freedom (look at column 3 of the above ANOVA table).

The final table shows computation of the test statistic which can be computed as follows:

$$F_{cal} = \frac{MSR}{MSE} \approx F_{\alpha}(k-1, n-k) \quad [\text{The F statistic follows F distribution}]$$

*The*

*test*

*rule:*

$$\text{Reject } H_0 \text{ if } F_{cal} > F_{\alpha}(k-1, n-k)$$

where  $F_{\alpha}(k-1, n-k)$  is the

value to be read from the F- distribution table at a given

$\alpha$

level.

### Relationship between F and R<sup>2</sup>

You may recall that  $R^2$  is given by  $\frac{\sum y^2}{\sum y^2 + \sum e^2}$  and  $\sum y^2 = R^2 \sum y^2$

We also know that

Hence,  $\frac{\sum e^2}{\sum y^2} = 1 - R^2$  which means  $\sum e^2 = (1 - R^2) \sum y^2$

The formula for F is:

$$F_{\alpha} = \frac{\frac{R^2 \sum y^2}{k-1}}{\frac{(1-R^2) \sum y^2}{n-k}} = \frac{R^2 \sum y^2}{(1-R^2) \sum y^2} \cdot \frac{n-k}{k-1}$$

That means the calculated F can also be expressed in terms of the coefficient of determination.

### Testing the Equality of two Regression Coefficients

Given the multiple regression equation:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$$

We would like to test the hypothesis:

$$H_0 : \beta_1 = \beta_2$$

or

$$\beta_1 - \beta_2 = 0 \quad \text{vs.} \quad H_1 : H_0 \text{ is not true}$$

The null hypothesis says that the two slope coefficients are equal.

**Example:** If Y is quantity demanded of a commodity,  $X_1$  is the price of the commodity and  $X_2$  is income of the consumer. The hypothesis suggests that the price and income elasticity of demand are the same.

We can test the null hypothesis using the classical assumption that

$$t = \frac{\hat{\beta}_1 - \hat{\beta}_2}{SE(\hat{\beta}_1 - \hat{\beta}_2)} \text{ t distribution with } N - K \text{ degrees of freedom.}$$

Where K = the total number of parameters estimated.

The  $SE(\hat{\beta}_1 - \hat{\beta}_2)$  is given as  $SE(\hat{\beta}_1 - \hat{\beta}_2) = \sqrt{Var(\hat{\beta}_1) + Var(\hat{\beta}_2) - 2cov(\hat{\beta}_1, \hat{\beta}_2)}$

Thus the t-statistic is:

$$t = \frac{\hat{\beta}_1 - \hat{\beta}_2}{SE(\hat{\beta}_1 - \hat{\beta}_2)}$$

Decision: Reject  $H_0$  if  $t_{cal.} > t_{tab.}$

**Note:** Using similar procedures one can also test linear equality restrictions, for example  $\beta_1 + \beta_2 = 1$  and other restrictions.

**Illustration:** The following table shows a particular country's the value of imports (Y), the level of Gross National Product( $X_1$ ) measured in arbitrary units, and the price index of imported goods ( $X_2$ ), over 12 years period.

Table 7: Data for multiple regression examples

| Year  | 1960 | 1961 | 1962 | 1963 | 1964 | 1965 | 1966 | 1967 | 1968 | 1969 | 1970 | 1971 |
|-------|------|------|------|------|------|------|------|------|------|------|------|------|
| Y     | 57   | 43   | 73   | 37   | 64   | 48   | 56   | 50   | 39   | 43   | 69   | 60   |
| $X_1$ | 220  | 215  | 250  | 241  | 305  | 258  | 354  | 321  | 370  | 375  | 385  | 385  |
| $X_2$ | 125  | 147  | 118  | 160  | 128  | 149  | 145  | 150  | 140  | 115  | 155  | 152  |

a) Estimate the coefficients of the economic relationship and fit the model.

To estimate the coefficients of the economic relationship, we compute the entries given in Table 8

**Table 8: Computations of the summary statistics for coefficients for data of Table 7**

| <b>Year</b> | <b>Y</b>     | <b>X<sub>1</sub></b> | <b>X<sub>2</sub></b> | <b>x<sub>1</sub></b> | <b>x<sub>2</sub></b> | <b>Y</b> | <b>X<sub>1</sub><sup>2</sup></b> | <b>x<sub>2</sub><sup>2</sup></b> | <b>x<sub>1</sub>y</b> | <b>x<sub>2</sub>y</b> | <b>x<sub>1</sub>x<sub>2</sub></b> | <b>y<sup>2</sup></b> |
|-------------|--------------|----------------------|----------------------|----------------------|----------------------|----------|----------------------------------|----------------------------------|-----------------------|-----------------------|-----------------------------------|----------------------|
| 1960        | 57           | 220                  | 125                  | -86.5833             | -15.3333             | 3.75     | 7496.668                         | 235.1101                         | -324.687              | -57.4999              | 1327.608                          | 14.0625              |
| 1961        | 43           | 215                  | 147                  | -91.5833             | 6.6667               | -10.25   | 8387.501                         | 44.44489                         | 938.7288              | -68.3337              | -610.558                          | 105.0625             |
| 1962        | 73           | 250                  | 118                  | -56.5833             | -22.3333             | 19.75    | 3201.67                          | 498.7763                         | -1117.52              | -441.083              | 1263.692                          | 390.0625             |
| 1963        | 37           | 241                  | 160                  | -65.5833             | 19.6667              | -16.25   | 4301.169                         | 386.7791                         | 1065.729              | -319.584              | -1289.81                          | 264.0625             |
| 1964        | 64           | 305                  | 128                  | -1.5833              | -12.3333             | 10.75    | 2.506839                         | 152.1103                         | -17.0205              | -132.583              | 19.52731                          | 115.5625             |
| 1965        | 48           | 258                  | 149                  | -48.5833             | 8.6667               | -5.25    | 2360.337                         | 75.11169                         | 255.0623              | -45.5002              | -421.057                          | 27.5625              |
| 1966        | 56           | 354                  | 145                  | 47.4167              | 4.6667               | 2.75     | 2248.343                         | 21.77809                         | 130.3959              | 12.83343              | 221.2795                          | 7.5625               |
| 1967        | 50           | 321                  | 150                  | 14.4167              | 9.6667               | -3.25    | 207.8412                         | 93.44509                         | -46.8543              | -31.4168              | 139.3619                          | 10.5625              |
| 1968        | 39           | 370                  | 140                  | 63.4167              | -0.3333              | -14.25   | 4021.678                         | 0.111089                         | -903.688              | 4.749525              | -21.1368                          | 203.0625             |
| 1969        | 43           | 375                  | 115                  | 68.4167              | -25.3333             | -10.25   | 4680.845                         | 641.7761                         | -701.271              | 259.6663              | -1733.22                          | 105.0625             |
| 1970        | 69           | 385                  | 155                  | 78.4167              | 14.6667              | 15.75    | 6149.179                         | 215.1121                         | 1235.063              | 231.0005              | 1150.114                          | 248.0625             |
| 1971        | 60           | 385                  | 152                  | 78.4167              | 11.6667              | 6.75     | 6149.179                         | 136.1119                         | 529.3127              | 78.75022              | 914.8641                          | 45.5625              |
| Sum         | 639          | 3679                 | 1684                 | 0.0004               | 0.0004               | 0        | 49206.92                         | 2500.667                         | 1043.25               | -509                  | 960.6667                          | 1536.25              |
| <b>Mean</b> | <b>53.25</b> | <b>306.5833</b>      | <b>140.3333</b>      | <b>0</b>             | <b>0</b>             | <b>0</b> |                                  |                                  |                       |                       |                                   |                      |

From Table 8, we can take the following summary results.

Table 8

Table 8

Table 8

$$\bar{X}_2 = \frac{\sum X_2}{n} = \frac{1684}{12} = 140.3333$$

The summary results in deviation forms are then given by:

Table 8

Table 8

$$\sum x_1 y = 1043.25$$

$$\sum x_2 y = -509$$

$$\sum x_1 x_2 = 960.6667$$

$$\sum y^2 = 1536.25$$

The coefficients are then obtained as follows.

Table 8

$$\hat{\beta}_1 = \frac{\sum x_1 y}{\sum x_1^2} = \frac{1043.25}{12212724} = 0.025365$$

The fitted model is then written as:  $\hat{Y}_i = 75.40512 + 0.025365X_1 - 0.21329X_2$

b) Compute the variance and standard errors of the slopes.

First, you need to compute the estimate of the variance of the random term as follows

$$\hat{\sigma}_e^2 = \frac{\sum e_i^2}{n-3} = \frac{1401.223}{12-3} = \frac{1401.223}{9} = 155.69143$$

Variance of  $\hat{\beta}_1$

$$Var(\hat{\beta}_1) = \hat{\sigma}_e^2 \left[ \frac{\sum x_2^2}{\sum x_1^2 \sum x_2^2 - (\sum x_1 x_2)^2} \right] = 155.69143 \left( \frac{2500.667}{12212724} \right) = 0.003188$$

Standard error of

$\hat{\beta}_1$

$$SE(\hat{\beta}_1) = \sqrt{Var(\hat{\beta}_1)} = \sqrt{0.003188} = 0.056462$$

Variance of  $\hat{\beta}_2$

$$Var(\hat{\beta}_2) = \hat{\sigma}_e^2 \left[ \frac{\sum x_1^2}{\sum x_1^2 \sum x_2^2 - (\sum x_1 x_2)^2} \right] = 155.69143 \left( \frac{49206.92}{12212724} \right) = 0.0627$$

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Standard error of

$\hat{\beta}_2$

$$SE(\hat{\beta}_2) = \sqrt{\text{Var}(\hat{\beta}_2)} = \sqrt{0.0627} = 0.25046$$

Similarly, the standard error of the intercept is found to be 37.98177. The detail is left for you as an exercise.

c) Calculate and interpret the coefficient of determination.

We can use the following summary results to obtain the  $R^2$ .

$$\sum \hat{y}^2 = 135.0262$$

$$\sum e^2 = 1401.223$$

$$\sum y^2 = 1536.25 \quad (\text{The sum of the above two}). \text{ Then,}$$

$$R^2 = \frac{\sum \hat{y}^2}{\sum y^2} = \frac{135.0262}{1536.25} = 0.087894$$

or

$$R^2 = 1 - \frac{\sum e^2}{\sum y^2} = 1 - \frac{1401.223}{1536.25} = 0.087894$$

d) Compute the adjusted  $R^2$ .

$$R_a^2 = 1 - \frac{(1-R^2)(n+1)}{n-2} = 1 - \frac{(1-0.087894)(9+1)}{9-2} = 0.0000$$

e) Construct 95% confidence interval for the true population parameters (partial regression coefficients). [Exercise: Base your work on Simple Linear Regression]

f) Test the significance of  $X_1$  and  $X_2$  in determining the changes in  $Y$  using  $t$ -test.

The hypotheses are summarized in the following table.

| Coefficient | Hypothesis                                | Estimate | Std. error | Calculated t                         | Conclusion                                       |
|-------------|-------------------------------------------|----------|------------|--------------------------------------|--------------------------------------------------|
| $\beta_1$   | $H_0: \beta_1=0$<br>$H_1: \beta_1 \neq 0$ | 0.025365 | 0.056462   | $\frac{0.025365}{0.056462} = 0.4494$ | We do not reject $H_0$ since $t_{cal} < t_{tab}$ |
| $\beta_2$   | $H_0: \beta_2=0$<br>$H_1: \beta_2 \neq 0$ | -0.21329 | 0.25046    | $\frac{-0.21329}{0.25046} = -0.8516$ | We do not reject $H_0$ since $t_{cal} < t_{tab}$ |

The critical value ( $t_{0.05, 9}$ ) to be used here is 2.262. Like the standard error test, the  $t$ -test revealed that both  $X_1$  and  $X_2$  are insignificant to determine ~~the~~ change in  $Y$  since the calculated  $t$  values are both less than the critical value.

**Exercise:** Test the significance of  $X_1$  and  $X_2$  in determining the changes in  $Y$  using the standard error test.

g) Test the overall significance of the model. (Hint: use  $\alpha = 0.05$ )

This involves testing whether at least one of the two variables  $X_1$  and  $X_2$  determine the changes in  $Y$ . The hypothesis to be tested is given by:

The ANOVA table for the test is given as follows:

| Source of variation | Sum of Squares                    | Degrees of freedom        | Mean Sum of Squares                                                  | $F_{cal}$                                                 |
|---------------------|-----------------------------------|---------------------------|----------------------------------------------------------------------|-----------------------------------------------------------|
| Regression          | $SSR = \sum \hat{y}^2 = 135.0262$ | $k - 1$<br>$= 3 - 1 = 2$  | $MSR = \frac{\sum \hat{y}^2}{k - 1} = \frac{135.0262}{2} = 67.51309$ | $F = \frac{MSR}{MSE} = \frac{67.51309}{155.614} = 0.4336$ |
| Residual            | $SSE = \sum e^2 = 1401.223$       | $n - k$<br>$= 12 - 3 = 9$ | $MSE = \frac{\sum e^2}{n - k} = \frac{1401.223}{9} = 155.614$        |                                                           |
| Total               | $\sum y^2 = 1536.25$              | $n - 1 = 12 - 1 = 11$     |                                                                      |                                                           |

The tabulated  $F$  value (critical value) is  $F_{\alpha}(2, 11) = 3.98$

In this case, the calculated  $F$  value (0.4336) is less than the tabulated value (3.98). Hence, we do not reject the null hypothesis and conclude that there is no significant contribution of the variables  $X_1$  and  $X_2$  to the changes in  $Y$ .

h) Compute the  $F$  value using the  $R^2$ .

$$F_{cal} = \frac{(n-k) R^2}{k(1-R^2)} = \frac{(12-3) \cdot 0.06794}{3 \cdot (1-0.06794)} = 0.4336$$

## Dummy Variable Regression Models

There are four basic types of variables we generally encounter in empirical analysis. These are: nominal, ordinal, interval and ratio scale variables. In preceding sections, we have encountered ratio scale variables. However, regression models do not deal only with ratio scale variables; they can also involve nominal and ordinal scale variables. In regression analysis, the dependent variable can be influenced by nominal variables such as sex, race, color, geographical region etc. models where all regressors are nominal (categorical) variables are called **ANOVA** (Analysis of Variance) models. If there is mixture of nominal and ratio scale variables, the models are called **ANCOVA** (Analysis of Covariance) models. Look at the following example.

**Illustration:** The following model represents the relationship between geographical location and teachers' average salary in public schools. The data were taken from 50 states for a single year. The 50 states were classified into three regions: Northeast, South and West. The regression model looks like the following.

$$Y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + u_i$$

Where  $Y_i$  = the (average) salary of public school teachers in state  $i$

$D_{1i} = 1$  if the state is in the Northeast

= 0 otherwise (i.e. in other regions of the country)

$D_{2i} = 1$  if the state is in the South

= 0 otherwise (i.e. in other regions of the country)

Note that the above regression model is like any multiple regression model considered previously, except that instead of quantitative regressors, we have only qualitative (dummy) regressors. Dummy regressors take value of 1 if the observation belongs to that particular category and 0 otherwise.

Note also that there are 3 states (categories) for which we have created only two dummy variables ( $D_1$  and  $D_2$ ). One of the rules in dummy variable regression is that if there are  $m$  categories, we need only  $m-1$  dummy variables. If we are suppressing the intercept, we can have  $m$  dummies but the interpretation will be a bit different.

The intercept value represents the **mean** value of the dependent variable for the benchmark category. This is the category for which we do not assign a dummy (in our case, West is a benchmark category). The coefficients of the dummy variable are called **differential intercept coefficients** because they tell us by how much the value of the intercept that receives the value of 1 differs from the intercept coefficient of the benchmark category.

$$\hat{Y} = 26158.62 - 1734.47 D_1 - 3264.42 D_2$$

$$Se = (1128.32) \quad (1453.95) \quad (1499.62)$$

$$F(2, 47) = 1.34$$

$$p\text{-value} \quad (0.000) \quad (0.233) \quad (0.0349) \quad R^2 = 0.0901$$

From the above fitted model, we can see that mean salary of public school teachers in the West is about \$26,158.62. The mean salary of teachers in the Northeast is lower by \$1734.47 than those of the West and those in the South is lower by \$3264.42. Doing this, we will find the average salaries in the latter two regions are about \$24,424 and \$22,894.

In order to know the statistical significance of the mean salary differences, we can run the tests we have discussed in previous sections. The other results can also be interpreted the way we discussed previously.

## Extensions of Regression Models

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As pointed out earlier non linearity may be expected in many Economic Relationships. In other words the relationship between  $Y$  and  $X$  can be non-linear rather than linear. Thus, once the independent variables have been identified the next step is to choose the functional form

of the relationship between the dependent and the independent variables. Specification of the functional form is important, because a correct explanatory variable may well appear to be insignificant or to have an unexpected sign if an inappropriate functional form is used. Thus the choice of a functional form for an equation is a vital part of the specification of that equation. The choice of a functional form almost always should be based on an examination of the underlying economic theory. The logical form of the relationship between the dependent variable and the independent variable in question should be compared with the properties of various functional forms, and the one that comes closest to that underlying theory should be chosen for the equation.

### Some Commonly Used Functional Forms

- a) *The Linear Form*: It is based on the assumption that the slope of the relationship between the independent variable and the dependent variable is constant.

$$\frac{\partial Y}{\partial X} = \beta_i \quad i=1,2,\dots,K$$



In this case elasticity is not constant.

$$\frac{\partial Y}{\partial X} = \beta_i$$

If the hypothesized relationship between Y and X is such that the slope of the relationship can be expected to be constant and the elasticity can therefore be expected to be variable, then the linear functional form should be used.

**Note:** Economic theory frequently predicts only the sign of a relationship and not its functional form. Under such circumstances, the linear form can be used until strong evidence that it is inappropriate is found. Thus, unless theory, common sense, or experience justifies using some other functional form, one should use the linear model.

- b) *Log-linear, double Log or constant elasticity model*

The most common functional form that is non-linear in the variable (but still linear in the coefficients) is the log-linear form. A log-linear form is often used, because the elasticities and not the slopes are constant i.e.,  $\eta = \beta$  Constant.



Thus, given the assumption of a constant elasticity, the proper form is the exponential (log-linear) form.

Given: 
$$Y = \beta_0 X^{\beta_1}$$

The log-linear functional form for the above equation can be obtained by a logarithmic transformation of the equation.

$$\ln Y = \ln \beta_0 + \beta_1 \ln X$$

The model can be estimated by OLS if the basic assumptions are fulfilled.



The model is also called a constant elasticity model because the coefficient of elasticity between Y and X ( $\beta_1$ ) remains constant.

$$\frac{\partial Y}{\partial X} = \beta_1$$

This functional form is used in the estimation of demand and production functions.

**Note:** We should make sure that there are no negative or zero observations in the data set before we decide to use the log-linear model. Thus log-linear models should be run only if all the variables take on positive values.

#### c) *Semi-log Form*

The semi-log functional form is a variant of the log-linear equation in which some but not all of the variables (dependent and independent) are expressed in terms of their logs. Such models expressed as:

( i )  $Y_i = \beta_0 + \beta_1 \ln X_{ii} + U_i$  ( lin-log model ) and ( ii )  $\ln Y_i = \beta_0 + \beta_1 \ln X_{ii} + U_i$  ( log-lin model ) are called semi-log models. The semi-log functional form, in the case of taking the log of one of the independent variables, can be used to depict a situation in which the impact of X on Y is expected to tail off as X gets bigger as long as  $\beta_1$  is greater than zero.



Example: The Engel's curve tends to flatten out, because as incomes get higher, a smaller percentage of income goes to consumption and a greater percentage goes to saving.

⇒ Consumption thus increases at a decreasing rate.

⇒ Growth models are examples of semi-log forms

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#### d) *Polynomial Form*

Polynomial functional forms express Y as a function of independent variables some of which are raised to powers others than one. For example in a second degree polynomial (quadratic) equation, at least one independent variable is squared.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_1^2 + \beta_3 X_2 + U_i$$

Such models produce slopes that change as the independent variables change. Thus the slopes of Y with respect to the Xs are

$$\frac{\Delta Y}{\Delta X_1} = \beta_1 + 2\beta_2 X_1, \text{ and } \frac{\Delta Y}{\Delta X_2} = \beta_3$$

In most cost functions, the slope of the cost curve changes as output changes.



Simple transformation of the polynomial could enable us to use the OLS method to estimate the parameters of the model

Setting  $X_3 = X_1^2$

$$\Rightarrow Y = \beta_0 + \beta_1 X_{1i} + \beta_2 X_3 + \beta_3 X_{2i} + U_i$$

#### e) Reciprocal Transformation (Inverse Functional Forms)

The inverse functional form expresses Y as a function of the reciprocal (or inverse) of one or more of the independent variables (in this case  $X_1$ ):

$$Y_i = \beta_0 + \beta_1 \left( \frac{1}{X_{1i}} \right) + \beta_2 X_{2i} + U_i$$

Or

$$Y_i = \beta_0 + \beta_1 \left( \frac{1}{\sqrt{X_{1i}}} \right) + \beta_2 X_{2i} + U_i$$

The reciprocal form should be used when the impact of a particular independent variable is expected to approach zero as that independent variable increases and eventually approaches infinity. Thus as  $X_1$  gets larger, its impact on Y decreases.

$$\frac{\Delta Y}{\Delta X_1}$$

An asymptote or limit value is set that the dependent variable will take if the value of the X-variable increases indefinitely i.e.  $\beta_0$  provides the value in the above case. The function approaches the asymptote from the top or bottom depending on the sign of  $\beta_1$ .

Example: Phillips curve, a non-linear relationship between the rate of unemployment and the percentage wage change.

$$W_t = \beta_0 + \beta_1 \left( \frac{1}{U_t} \right) + U_t$$

### Summary

Multiple regression model is an extension of the two variable regression model with new concepts involved and more practicality. Such models can be used for the purpose of mean and individual prediction. This is the simplest possible multiple linear regression model is the three variable regression model.  $R^2$  and adjusted  $R^2$  are overall measures of how well the chosen model fits the data. Hypothesis testing in multiple linear regression models include testing the individual statistical significance of partial regression coefficients, testing overall significance of model and others. Dummy variables classify set of samples into subgroups based on qualities or attributes. It is common to see regression models with dummy variables as regressors. The coefficients should be interpreted very carefully.

### Statistical Software Application Applications and Practices

## CHAPTER SIX

### ECONOMETRIC PROBLEMS

#### Pre-test Questions

1. What are the major CLRM assumptions?
2. What happens to the properties of the OLS estimators if one or more of these assumptions are violated, i.e. not fulfilled?
3. How we can check if an assumption is violated or not?

#### Assumptions Revisited

In many practical cases, two major problems arise in applying the classical linear regression model.

- 1) those due to assumptions about the specification of the model and about the disturbances and
- 2) those due to assumptions about the data

The following assumptions fall in either of the categories.

- The regression model is linear in parameters.
- The values of the explanatory variables are fixed in repeated sampling (non-stochastic).
- The mean of the disturbance ( $u_i$ ) is zero for any given value of  $X$  i.e.  $E(u_i) = 0$
- The variance of  $u_i$  is constant i.e. homoscedastic
- There is no autocorrelation in the disturbance terms
- The explanatory variables are independently distributed with the  $u_i$ .
- The number of observations must be greater than the number of explanatory variables.
- There must be sufficient variability in the values taken by the explanatory variables.
- There is no linear relationship (multicollinearity) among the explanatory variables.
- The stochastic (disturbance) term  $u_i$  are normally distributed i.e.,  $u_i \sim N(0, \sigma^2)$
- The regression model is correctly specified i.e., no specification error.

With these assumptions we can show that OLS are BLUE, and normally distributed. Hence it was possible to test Hypothesis about the parameters. However, if any of such assumption is relaxed, the OLS might not work. We shall not examine in detail the violation of some of the assumptions.

#### Violations of Assumptions

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#### The Zero Mean Assumption i.e. $E(u_i)=0$

If this assumption is violated, we obtain a biased estimate of the intercept term. But, since the intercept term is not very important we can leave it. The slope coefficients remain unaffected even if the assumption is violated. The intercept term does not also have physical interpretation.

### **The Normality Assumption**

This assumption is not very essential if the objective is estimation only. The OLS estimators are BLUE regardless of whether the  $u_i$  are normally distributed or not. In addition, because of the central limit theorem, we can argue that the test procedures – the t-tests and F-tests – are still valid asymptotically, i.e. in large sample.

### **Heteroscedasticity: The Error Variance is not Constant**

The error terms in the regression equation have a common variance i.e., are Homoscedastic. If they do not have common variance we say they are **Heteroscedastic**. The basic questions to be addressed are:

- What is the nature of the problem?
- What are the consequences of the problem?
- How do we detect (diagnose) the problem?
- What remedies are available for the problem?

### **The Nature of the Problem**

In the case of homoscedastic disturbance terms the spread around the mean is constant, i.e.  $= \sigma^2$ . But in the case of heteroscedasticity disturbance terms the variance changes with the explanatory variable. The problem of heteroscedasticity is likely to be more common in cross-sectional than in time-series data.

### **Causes of Heteroscedasticity**

There are several reasons why the variance of the error term may be variable, some of which are as follows.

- Following the *error-learning* models, as people learn, their errors of behaviour become smaller over time where the standard error of the regression model decreases.
- As income grows people have discretionary income and hence more scope for choice about the disposition of their income. Hence, the variance (standard error) of the regression is more likely to increase with income.
- Improvement in data collection techniques will reduce errors (variance).
- Existence of outliers might also cause heteroscedasticity.
- Misspecification of a model can also be a cause for heteroscedasticity.
- Skewness in the distribution of one or more explanatory variables included in the model is another source of heteroscedasticity.
- Incorrect data transformation and incorrect functional form are also other sources

**Note:** Heteroscedasticity is likely to be more common in cross-sectional data than in time series data. In cross-sectional data, individuals usually deal with samples (such as consumers, producers, etc) taken from a population at a given point in time. Such members might be of different size. In time series data, however, the variables tend to be of similar orders of magnitude since data is collected over a period of time.

### Consequences of Heteroscedasticity

If the error terms of an equation are heteroscedastic, there are three major consequences.

- a) The ordinary least square estimators are still linear since heteroscedasticity does not cause bias in the coefficient estimates. The least square estimators are still unbiased.
- b) Heteroscedasticity increases the variance of the partial regression coefficients but it does affect the minimum variance property. Thus, the OLS estimators are inefficient. Thus the test statistics t-test and F-test cannot be relied on in the face of uncorrected heteroscedasticity.

### Detection of Heteroscedasticity

There are no hard and fast rules (universally agreed upon methods) for detecting the presence of heteroscedasticity. But some rules of thumb can be suggested. Most of these methods are based on the examination of the OLS residuals,  $e_i$ , since these are the once we observe and not the disturbance term  $u_i$ . There are informal and formal methods of detecting heteroscedasticity.

#### a) Nature of the problem

In cross-sectional studies involving heterogeneous units, heteroscedasticity is the rule rather than the exception.

**Example:** In small, medium and large sized agribusiness firms in a study of input expenditure in relation to sales, the rate of interest, etc. heteroscedasticity is expected.

#### b) Graphical method

If there is no a priori or empirical information about the nature of heteroscedasticity, one could do an examination of the estimated residual squared,  $e_i^2$  to see if they exhibit any systematic pattern. The squared residuals can be plotted either against Y or against one of the explanatory variables. If there appears any systematic pattern, heteroscedasticity might exist. These two methods are informal methods.

#### c) Park Test

Park suggested a statistical test for heteroscedasticity based on the assumption that the variance of the disturbance term ( $\sigma_i^2$ ) is some function of the explanatory variable  $X_i$ .

Park suggested a functional form as:

$$\sigma_i^2 = \sigma^2 X_i^\beta e^{\nu_i}$$

which can be transferred to a linear function using ln transformation. Hence,

$Var(e_i) = \sigma^2 X_i^{\beta_i} e^{v_i}$  where  $v_i$  is the stochastic disturbance term.

$$\ln \sigma_i^2 = \ln \sigma^2 + \beta \ln X_i + v_i$$

since  $\sigma^2$  is not known.

The Park-test is a two-stage procedure: run OLS regression disregarding the heteroscedasticity question and obtain the  $e_i$  and then run the above equation. The regression is run and if  $\beta$  turns out to be statistically significant, then it would suggest that heteroscedasticity is present in the data.

#### d) Spearman's Rank Correlation test

Recall that:  $r_s = 1 - 6 \left[ \frac{\sum d_i^2}{N(N^2 - 1)} \right]$  d = difference between ranks

**Step 1:** Regress Y on X and obtain the residuals,  $e_i$ .

**Step 2:** Ignoring the significance of  $e_i$  or taking  $|e_i|$  rank both  $e_i$  and  $X_i$  and compute the rank correlation coefficient.

**Step 3:** Test for the significance of the correlation statistic by the t-test

$$t = \frac{r_s \sqrt{N-2}}{\sqrt{1-r_s^2}} \sim t(n-2)$$

A high rank correlation suggests the presence of heteroscedasticity. If more than one explanatory variable, compute the rank correlation coefficient between  $e_i$  and each explanatory variable separately.

#### e) Goldfeld and Quandt Test

This is the most popular test and usually suitable for large samples. If it is assumed that the variance ( $\sigma_i^2$ ) is positively related to one of the explanatory variables in the regression model and if the number of observations is at least twice as many as the parameters to be estimated, the test can be used.

Given the model

$$Y_i = \beta_0 + \beta_1 X_i + U_i$$

Suppose  $\sigma_i^2$  is positively related to  $X_i$  as

$$\sigma_i^2 = \sigma^2 X_i^2$$

Goldfeld and Quandt suggest the following steps:

1. Rank the observation according to the values of  $X_i$  in ascending order.

2. Omit the central  $c$  observations (usually the middle third of the recorded observations), or where  $c$  is specified a priori, and divide the remaining  $(n-c)$  observations into two groups, each with  $\frac{(n-c)}{2}$  observations.
3. Fit separate regressions for the two sub-samples and obtain the respective residuals  $RSS_1$  and  $RSS_2$  with  $\frac{(n-c)}{2} - k$  df
4. Compute the ratio:
 
$$F = \frac{\frac{1}{n-c-k} \sum_{i=1}^{n-c-k} e_i^2}{\frac{1}{n-c-k} \sum_{i=1}^{n-c-k} e_i^2} = \frac{(n-c-2k)}{2}$$

If the two variances tend to be the same, then  $F$  approaches unity. If the variances differ we will have values for  $F$  different from one. The higher the  $F$ -ratio, the stronger the evidence of heteroscedasticity.

**Note:** There are also other methods of testing the existence of heteroscedasticity in your data. These are Glejser Test, Breusch-Pagan-Godfrey Test, White's General Test and Koenker-Bassett Test the details for which you are supposed to refer.

### Remedial Measures

OLS estimators are still unbiased even in the presence of heteroscedasticity. But they are not efficient, not even asymptotically. This lack of efficiency makes the usual hypothesis testing procedure a dubious exercise. Remedial measures are, therefore, necessary. Generally the solution is based on some form of transformation.

#### a) The Weighted Least Squares (WLS)

Given a regression equation model of the form

$$Y_i = \beta_0 + \beta_1 X_i + U_i$$

The weighted least square method requires running the OLS regression to a transformed data. The transformation is based on the assumption of the form of heteroscedasticity.

**Assumption One:** Given the model  $Y_i = \beta_0 + \beta_1 X_i + U_i$

If  $\text{var}(U_i) = \sigma_i^2 = \sigma^2 X_i^2$ , then  $E(U_i) = \sigma^2 X_i^2$

Where  $\sigma^2$  is a constant variance of a classical error term. So if as a matter of speculation or other tests indicate that the variance is proportional to the square of the explanatory variable  $X$ , we may transform the original model as follows:

$$\frac{Y_i}{X_i^2} = \frac{\beta_0}{X_i^2} + \frac{\beta_1 X_i}{X_i^2} + \frac{U_i}{X_i^2}$$

$$\text{Now } E\left(\frac{Y_i}{X_i^2}\right) = E\left(\frac{U_i}{X_i^2}\right) = \frac{1}{X_i^2} E(U_i^2) = \sigma^2$$

Hence the variance of  $U_i$  is now homoscedastic and regress  $\frac{Y_i}{X_i^2}$  on  $\frac{1}{X_i^2}$ .

**Assumption Two:** Again given the model

$$Y_i = \beta_0 + \beta_1 X_i + U_i$$

Suppose now  $Var(U_i) = E(U_i^2) = \sigma^2 X_{1i}$

It is believed that the variance of  $U_i$  is proportional to  $X_{1i}$ , instead of being proportional to the squared  $X_{1i}$ . The original model can be transformed as follows:

$$\begin{aligned} \frac{Y_i}{\sqrt{X_{1i}}} &= \frac{\beta_0}{\sqrt{X_{1i}}} + \beta_1 \frac{X_{1i}}{\sqrt{X_{1i}}} + \frac{U_i}{\sqrt{X_{1i}}} \\ &= \beta_0 \frac{1}{\sqrt{X_{1i}}} + \beta_1 \sqrt{X_{1i}} + V_i \end{aligned}$$

Where  $V_i = U_i / \sqrt{X_{1i}}$  and where  $X_{1i} > 0$

Thus,  $E(U_i^2) = E\left[\left(\frac{Y_i}{\sqrt{X_{1i}}} - \frac{\beta_0}{\sqrt{X_{1i}}} - \beta_1 \sqrt{X_{1i}}\right)^2\right] = \sigma^2 X_{1i}$

Now since the variance of  $V_i$  is constant (homoscedastic) one can apply the OLS technique to

regress  $\frac{Y_i}{\sqrt{X_{1i}}}$  on  $\frac{1}{\sqrt{X_{1i}}}$  and  $\sqrt{X_{1i}}$ .

To go back to the original model, one can simply multiply the transformed model by  $\sqrt{X_{1i}}$ .

*Assumption Three:* Given the model let us assume that

$$E(U_i^2) = \sigma^2 [E(Y_i)]^2$$

The variance is proportional to the square of the expected value of  $Y$ .

Now,  $E(U_i^2) = \sigma^2 X_{1i}$

We can transform the original model as

$$\frac{Y_i}{\sqrt{X_{1i}}} = \frac{\beta_0}{\sqrt{X_{1i}}} + \beta_1 \sqrt{X_{1i}} + V_i$$

Again it can be verified that  $E\left(\frac{Y_i}{\sqrt{X_{1i}}}\right)^2$  gives us a constant variance  $\sigma^2$

$$E\left(\frac{Y_i}{\sqrt{X_{1i}}}\right)^2 = \frac{1}{X_{1i}} E(Y_i^2) = \frac{1}{X_{1i}} (\beta_0^2 + \beta_1^2 X_{1i} + \sigma^2 X_{1i}) = \beta_0^2 + \beta_1^2 + \sigma^2$$

The disturbance  $V_i$  is homoscedastic and the regression can be run.

*Assumption Four:* If instead of running the regression  $\frac{Y_i}{\sqrt{X_{1i}}}$  one could run

$$\ln Y_i = \beta_0 + \beta_1 \ln X_{1i} + U_i$$

Then it reduces heteroscedasticity.

### b) Other Remedies for Heteroscedasticity

Two other approaches could be adopted to remove the effect of heteroscedasticity.

- Include a previously omitted variable(s) if heteroscedasticity is suspected due to omission of variables.
- Redefine the variables in such a way that avoids heteroscedasticity. For example, instead of total income, we can use Income per capita.

### Autocorrelation: Error Terms are correlated

Another assumption of the regression model was the non-existence of serial correlation (autocorrelation) between the disturbance terms,  $U_i$ .

$$Cov(U_i, U_j) = 0 \quad i \neq j$$

Serial correlation implies that the error term from one time period depends in some systematic way on error terms from other time periods. Autocorrelation is more a problem of time series data than cross-sectional data. If by chance, such a correlation is observed in cross-sectional units, it is called *spatial autocorrelation*. So, it is important to understand serial correlation and its consequences of the OLS estimators.

### Nature of Autocorrelation

The classical model assumes that the disturbance term relating to any observation is not influenced by the disturbance term relating to any other disturbance term.

$$E(U_i U_j) = 0, i \neq j$$

But if there is any interdependence between the disturbance terms then we have autocorrelation

$$E(U_i U_j) \neq 0, i \neq j$$

### Causes of Autocorrelation

Serial correlation may occur because of a number of reasons.

- Inertia (built in momentum) a salient feature of most economic variables time series (such as GDP, GNP, price indices, production, employment etc) is inertia or sluggishness. Such variables exhibit (business) cycles.
- Specification bias exclusion of important variables or incorrect functional forms
- Lags in a time series regression, value of a variable for a certain period depends on the variable's previous period value.
- Manipulation of data if the raw data is manipulated (extrapolated or interpolated), autocorrelation might result.

Autocorrelation can be negative as well as positive. The most common kind of serial correlation is the **first order serial** correlation. This is the case in which this period error terms are functions of the previous time period error term.

$$E_t = \rho E_{t-1} + U_t$$

This is also called the first order autoregressive model.

$$-1 < \rho < 1$$

The disturbance term  $U_t$  satisfies all the basic assumptions of the classical linear model.

$$E(U_t) = 0$$

$$E(U_t, U_{t-1}) = 0 \quad t \neq t-1$$

$$U_t \sim N(0, \sigma^2)$$

### Consequences of serial correlation

When the disturbance term exhibits serial correlation, the values as well as the standard errors of the parameters are affected.

- 1) The estimates of the parameters remain unbiased even in the presence of autocorrelation but the  $X$ 's and the  $u$ 's must be uncorrelated.

- 2) Serial correlation increases the variance of the OLS estimators. The minimum variance property of the OLS parameter estimates is violated. That means the OLS are no longer efficient.
- 3) Due to serial correlation the variance of the disturbance term,  $U_i$  may be underestimated. This problem is particularly pronounced when there is positive autocorrelation.
- 4) If the  $U_i$ s are autocorrelated, then prediction based on the ordinary least squares estimates will be inefficient. This is because of larger variance of the parameters. Since the variances of the OLS estimators are not minimal as compared with other estimators, the standard error of the forecast from the OLS will not have the least value.

### Detecting Autocorrelation

Some rough idea about the existence of autocorrelation may be gained by plotting the residuals either against their own lagged values or against time.



Figure 3: Graphical detection of autocorrelation

There are more accurate tests for the incidence of autocorrelation. The most common test of autocorrelation is the Durbin-Watson Test.

### The Durbin-Watson d Test

The test for serial correlation that is most widely used is the Durbin-Watson d test. This test is appropriate only for the first order autoregressive scheme.

$$U_t = \rho U_{t-1} + E_t \text{ then } E_t = \rho E_{t-1} + U_t$$

The test may be outlined as

$$H_0 : \rho = 0$$

$$H_1 : \rho \neq 0$$

This test is, however, applicable where the underlying assumptions<sup>63</sup> are met:

- The regression model includes an intercept term
- The serial correlation is first order in nature

- The regression does not include the lagged dependent variable as an explanatory variable
- There are no missing observations in the data

The equation for the Durban-Watson d statistic is

$$d = \frac{\sum_{t=2}^n (e_t - e_{t-1})^2}{\sum_{t=1}^n e_t^2}$$

Which is simply the ratio of the sum of squared differences in successive residuals to the RSS. Note that the numerator has one fewer observation than the denominator, because an observation must be used to calculate  $e_t - e_{t-1}$ . A great advantage of the d-statistic is that it is based on the estimated residuals. Thus it is often reported together with  $R^2$ , t, etc.

The d-statistic equals **zero** if there is extreme positive serial correlation, **two** if there is no serial correlation, and **four** if there is extreme negative correlation.

1. Extreme positive serial correlation:  $d \approx 0$

$$e_t = e_{t-1} \text{ so } (e_t - e_{t-1}) \approx 0 \text{ and } d \approx 0.$$

2. Extreme negative correlation:  $d \approx 4$

$$e_t = -e_{t-1} \text{ and } d = \frac{\sum (2e_t)^2}{\sum e_t^2} \text{ and } d \approx 4$$

3. No serial correlation:  $d \approx 2$

$$d = \frac{\sum (e_t - e_{t-1})^2}{\sum e_t^2} = \frac{\sum e_t^2 + \sum e_{t-1}^2 - 2\sum e_t e_{t-1}}{\sum e_t^2} = 2$$

Since  $\sum e_t e_{t-1} = 0$ , because they are uncorrelated. Since  $\sum e_t^2$  and  $\sum e_{t-1}^2$  differ in only one observation, they are approximately equal.

The exact sampling or probability distribution of the d-statistic is not known and, therefore, unlike the t,  $X^2$  or F-tests there are no unique critical values which will lead to the acceptance or rejection of the null hypothesis.

But Durbin and Watson have successfully derived the upper and lower bound so that if the computed value d lies outside these critical values, a decision can be made regarding the presence of a positive or negative serial autocorrelation.

Thus

$$d = \frac{\sum (e_t - e_{t-1})^2}{\sum e_t^2} = \frac{\sum e_t^2 + \sum e_{t-1}^2 - 2\sum e_t e_{t-1}}{\sum e_t^2} = 2(1 - \frac{\sum e_t e_{t-1}}{\sum e_t^2})$$

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$$d = 2(1 - \hat{P}) \text{ since } \frac{\sum e_t e_{t-1}}{\sum e_t^2} = \hat{P}$$

But, since  $-1 \leq P \leq 1$  the above identity can be written as:  $0 \leq d \leq 4$

Therefore, the bounds of  $d$  must lie within these limits.



- Thus
- if  $\hat{\rho} = 0 \rightarrow d = 2$ , no serial autocorrelation.
  - if  $\hat{\rho} = +1 \rightarrow d = 0$ , evidence of positive autocorrelation.
  - if  $\hat{\rho} = -1 \rightarrow d = 4$ , evidence of negative autocorrelation.

#### **Decision Rules for Durbin-Watson - d-test**

| Null hypothesis             | Decision      | If                            |
|-----------------------------|---------------|-------------------------------|
| No positive autocorrelation | Reject        | $0 < d < d_L$                 |
| No positive autocorrelation | No decision   | $d_L \leq d \leq d_U$         |
| No negative autocorrelation | Reject        | $4 - d_L < d < 4$             |
| No negative autocorrelation | No decision   | $4 - d_U \leq d \leq 4 - d_L$ |
| No autocorrelation          | Do not reject | $d_U < d < 4 - d_U$           |

**Note:** Other tests for autocorrelation include the Runs test and the Breusch-Godfrey (BG) test. There are so many tests of autocorrelation since there is no particular test that has been judged to be unequivocally best or more powerful in the statistical sense.

#### **Remedial Measures for Autocorrelation**

Since in the presence of serial correlation the OLS estimators are inefficient, it is essential to seek remedial measures.

- 1) The solution depends on the source of the problem.
  - If the source is omitted variables, the appropriate solution is to include these variables in the set of explanatory variables.
  - If the source is misspecification of the mathematical form the relevant approach will be to change the form.
- 2) If these sources are ruled out then the appropriate procedure will be to transform the original data so as to produce a model whose random variable satisfies the assumptions of non-autocorrelation. But the transformation depends on the pattern of autoregressive structure. Here we deal with first order autoregressive scheme.

$$\varepsilon_t = \rho \varepsilon_{t-1} + U_t$$

For such a scheme the appropriate transformation is to subtract from the original observations of each period the product of  $\hat{\rho}$  times the value of the variables in the previous period.

where:

Thus, if the structure of autocorrelation is known, then it is possible to make the above transformation. But often the structure of the autocorrelation is not known. Thus, we need to estimate  $\mathbf{P}$  in order to make the transformation.

### When $\rho$ is not known

There are different ways of estimating the correlation coefficient,  $\rho$ , if it is unknown.

#### 1) Estimation of $\rho$ from the $d$ -statistic

Recall that  $d \approx 2(1-\hat{\rho})$  or  $\hat{\rho} \approx 1 - \frac{d}{2}$

which suggest a simple way of obtaining an estimate of  $\rho$  from the estimated  $d$  statistic. Once an estimate of  $\rho$  is made available one could proceed with the estimation of the OLS parameters by making the necessary transformation.

#### 2) Durbin's two step method

Given the original function as

$$U_t = \rho U_{t-1} + V_t$$

$$\text{let } U_t = \rho U_{t-1} + V_t$$

**Step 1:** start from the transformed model

$$U_t = \rho U_{t-1} + V_t$$

rearranging and setting

$$\begin{aligned} \beta_0(1-\rho) &= a_0 \\ \beta_1 &= a_1 \\ \beta_2\rho &= a_2 \\ \text{etc.} \end{aligned}$$

The above equation may be written as

$$[1-\rho]Y_t = [1-\rho]a_0 + \rho a_1 X_{t-1} + a_2 X_{t-2} + \dots + V_t$$

Applying OLS to the equation, we obtain an estimate of  $\rho$ , which is the coefficient of the lagged variable  $Y_{t-1}$ .

**Step 2:** We use this estimate,  $\hat{\rho}$  to obtain the transformed variables.

$$\begin{aligned} Y_t - \hat{\rho}Y_{t-1} &= Y_t^* \\ X_{t1} - \hat{\rho}X_{t-1,1} &= X_{t1}^* \\ \vdots \\ X_{tK} - \hat{\rho}X_{t-1,K} &= X_{tK}^* \end{aligned}$$

We use this model to estimate the parameters of the original relationship.

$$Y_t^* = \beta_0 + \beta_1 X_{t1}^* + \beta_2 X_{t2}^* + \dots + V_t$$

The methods discussed above to solve the problem of serial autocorrelation are basically two step methods. In step 1, we obtain an estimate of the unknown  $\rho$  and in step 2, we use that estimate to transform the variables to estimate the generalized difference equation.

**Note:** The Cochrane-Orcutt Iterative Method is also another method.

### Multicollinearity: Exact linear correlation between Regressors<sup>66</sup>

One of the classical assumptions of the regression model is that the explanatory variables are uncorrelated. If the assumption that no independent variable is a perfect linear function of one

or more other independent variables is violated we have the problem of multicollinearity. If the explanatory variables are perfectly linearly correlated, the parameters become indeterminate. It is impossible to find the numerical values for each parameter and the method of estimation breaks.

If the correlation coefficient is 0, the variables are called *orthogonal*; there is no problem of multicollinearity. Neither of the above two extreme cases is often met. But some degree of inter-correlation is expected among the explanatory variables, due to the interdependence of economic variables.

Multicollinearity is not a condition that either exists or does not exist in economic functions, but rather a phenomenon inherent in most relationships due to the nature of economic magnitude. But there is no conclusive evidence which suggests that a certain degree of multicollinearity will affect seriously the parameter estimates.

### Reasons for Existence of Multicollinearity

There is a tendency of economic variables to move together over time. For example, Income, consumption, savings, investment, prices, employment tend to rise in the period of economic expansion and decrease in a period of recession. The use of lagged values of some explanatory variables as separate independent factors in the relationship also causes multicollinearity problems.

**Example:**  $Consumption = f(Y_t, Y_{t-1}, \dots)$

Thus, it can be concluded that multicollinearity is expected in economic variables. Although multicollinearity is present also in cross-sectional data it is more a problem of time series data.

### Consequences of Multicollinearity

Recall that, if the assumptions of the classical linear regression model are satisfied, the OLS estimators of the regression estimators are BLUE. As stated above if there is perfect multicollinearity between the explanatory variables, then it is not possible to determine the regression coefficients and their standard errors. But if collinearity among the **X**-variables is high, but not perfect, then the following might be expected.

Nevertheless, the effect of collinearity is controversial and by no means conclusive.

- 1) The estimates of the coefficients are statistically unbiased. Even if an equation has significant multicollinearity, the estimates of the parameters will still be centered around the true population parameters.
- 2) When multicollinearity is present in a function, the variances and therefore the standard errors of the estimates will increase, although some econometricians argue that this is not always the case.

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- (3) The computed t-ratios will fall i.e. insignificant t-ratios will be observed in the presence of multicollinearity. 
$$t = \frac{\beta}{SE(\beta)}$$
 since

$SE(\hat{\beta})$  increases t-falls. Thus one may increasingly accept the null hypothesis that the relevant true population's value is zero

$$H_0 : \beta_i = 0$$

Thus because of the high variances of the estimates the null hypothesis would be accepted.

- (4) A high  $R^2$  but few significant t-ratios are expected in the presence of multicollinearity. So one or more of the partial slope coefficients are individually statistically insignificant on the basis of the t-test. Yet the  $R^2$  may be so high. Indeed, this is one of the signals of multicollinearity, insignificant t-values but a high overall  $R^2$  and F-values. Thus because multicollinearity has little effect on the overall fit of the equation, it will also have little effect on the use of that equation for prediction or forecasting.

### Detecting Multicollinearity

Having studied the nature of multicollinearity and the consequences of multicollinearity, the next question is how to detect multicollinearity. The main purpose in doing so is to decide how much multicollinearity exists in an equation, not whether any multicollinearity exists. So the important question is the degree of multicollinearity. But there is no one unique test that is universally accepted. Instead, we have some rules of thumb for assessing the severity and importance of multicollinearity in an equation. Some of the most commonly used approaches are the following:

#### 1) *High $R^2$ but few significant t-ratios*

This is the classical test or symptom of multicollinearity. Often if  $R^2$  is high ( $R^2 > 0.8$ ) the F-test in most cases will reject the hypothesis that the partial slope coefficients are simultaneously equal to zero, but the individual t-tests will show that none or very few partial slope coefficients are statistically different from zero. In other words, multicollinearity that is severe enough to substantially lower t-scores does very little to decrease  $R^2$  or the F-statistic.

So the combination of high  $R^2$  with low calculated t-values for the individual regression coefficients is an indicator of the possible presence of severe multicollinearity.

*Drawback:* a non-multicollinear explanatory variable may still have a significant coefficient even if there is multicollinearity between two or more other explanatory variables. Thus, equations with high levels of multicollinearity will often have one or two regression coefficients significantly different from zero, thus making the high  $R^2$  low t rule a poor indicator in such cases.

#### 1) *High pair-wise (simple) correlation coefficients among the regressors (explanatory variables).*

If the  $R$ 's are high in absolute value, then it is highly probable that the  $X$ 's are highly correlated and that multicollinearity is a potential problem. The question is how high  $r$  should be to suggest multicollinearity. Some suggest that if  $r$  is in excess of 0.80, then multicollinearity could be suspected.

Another rule of thumb is that multicollinearity is a potential problem when the squared simple correlation coefficient is greater than the unadjusted  $R^2$ .

Two  $X$ 's are severely multicollinear if  $(r_{x_j x_k})^2 \geq R^2$ .

A major problem of this approach is that although high zero-order correlations may suggest collinearity, it is not necessary that they be high to have collinearity in any specific case.

## 2) VIF and Tolerance

VIF shows the speed with which the variances and covariances increase. It also shows how the variance of an estimator is influenced by the presence of multicollinearity. VIF is defined as follows:

Where  $\frac{1}{1 - r_{12}^2}$  is the correlation between two explanatory variables. As  $r_{12}^2$  approaches 1, the VIF approaches infinity. If there no collinearity, VIF will be 1. As a rule of thumb, VIF value of 10 or more shows multicollinearity is sever problem. Tolerance is defined as the inverse of VIF.

## 3) Other more formal tests for multicollinearity

The use of formal tests to give any indications of the severity of the multicollinearity in a particular sample is controversial. Some econometricians reject even the simple indicators developed above, mainly because of the limitations cited. Some people tend to use a number of more formal tests. But none of these is accepted as the best.

## Remedies for Multicollinearity

There is no automatic answer to the question what can be done to minimize the problem of multicollinearity. The possible solution which might be adopted if multicollinearity exists in a function, vary depending on the severity of multicollinearity, on the availability of other data sources, on the importance of factors which are multicollinear, on the purpose for which the function is used. However, some alternative remedies could be suggested for reducing the effect of multicollinearity.

### 1) Do Nothing

Some writers have suggested that if multicollinearity does not seriously affect the estimates of the coefficients one may tolerate its presence in the function. In a sense, multicollinearity is similar to a non-life threatening human disease that requires an operation only if the disease is causing a significant problem. A remedy for multicollinearity should only be considered if and when the consequences cause insignificant t-scores or widely unreliable estimated coefficients.

### 2) Dropping one or more of the multicollinear variables

When faced with severe multicollinearity, one of the simplest way to get rid of (drop) one or more of the collinear variables. Since multicollinearity is caused by correlation between the explanatory variables, if the multicollinear variables are dropped the correlation no longer exists.

Some people argue that dropping a variable from the model may introduce specification error or specification biases. According to them since OLS estimators are still BLUE despite near collinearity omitting a variable may seriously mislead us as to the true values of the parameters.

**Example:** If economic theory says that income and wealth should both be included in the model explaining the consumption expenditure, dropping the wealth variable would constitute specification bias.

### 3) Transformation of the variables

If the variables involved are all extremely important on theoretical grounds, neither doing nothing nor dropping a variable could be helpful. But it is sometimes possible to transform the variables in the equation to get rid of at least some of the multicollinearity.

Two common such transformations are:

- (i) to form a linear combination of the multicollinear variables
- (ii) to transform the equation into first differences (or logs)

The technique of forming a linear combination of two or more of the multicollinearity variables consists of:

- creating a new variable that is a function of the multicollinear variables
- using the new variable to replace the old ones in the regression equation (if  $X_1$  and  $X_2$  are highly multicollinear, a new variable,  $X_3 = X_1 + X_2$  or  $K_1X_1 + K_2X_2$  might be substituted for both of the multicollinear variables in a re-estimation of the model)

The second kind of transformation to consider as possible remedy for severe multicollinearity is to change the functional form of the equation.

A first difference is nothing more than the change in a variable from the previous time-period.

$$\Delta X_t = X_t - X_{t-1}$$

If an equation (or some of the variables in an equation) is switched from its normal specification to a first difference specification, it is quite likely that the degree of multicollinearity will be significantly reduced for two reasons.

- Since multicollinearity is a sample phenomenon, any change in the definitions of the variables in that sample will change the degree of multicollinearity.
- Multicollinearity takes place most frequently in time-series data, in which first differences are far less likely to move steadily upward than are the aggregates from which they are calculated.

### (4) Increase the sample size

Another solution to reduce the degree of multicollinearity is to attempt to increase the size of the sample. Larger data set (often requiring new data collection) will allow more accurate

estimates than a small one, since the large sample normally will reduce somewhat the variance of the estimated coefficients reducing the impact of multicollinearity. But, for most economic and business applications, this solution is not feasible. As a result new data are generally impossible or quite expensive to find. One way to increase the sample is to pool cross-sectional and time series data.

### *3) Other Remedies*

There are several other methods suggested to reduce the degree of multicollinearity. Often multivariate statistical techniques such as Factor analysis and Principal component analysis or other techniques such as ridge regression are often employed to solve the problem of multicollinearity.

#### **Learning Activities and Software practices:**

Fitting Multiple Linear Regression model and check violations of assumptions of regression models, identify the causes and apply remedies to solve these problems.

#### **Summary**

The Ordinary Least Squares methods will work when the assumptions of classical linear regression models hold. One of the critical assumptions of the classical linear regression model is that the disturbances have all same variance the violation of which leads to heteroscedasticity. Heteroscedasticity does not destroy the unbiasedness and consistency properties of OLS estimators but the efficiency property. There are several diagnostic tests available for detecting it but one cannot tell for sure which will work in a given situation. Eventhough it is detected, it is not easy to correct it. Transforming the data might be a possible way out. The other assumption is that there is no multicollinearity (exact or approximately exact linear relationship) among the explanatory variables. If there is perfect collinearity, the regression coefficients are indeterminate. Although there are no sure methods of detecting collinearity, there are several indicators of it. The clearest sign of it is when  $R^2$  is very high but none of the regression coefficients is statistically significant. Detection of multicollinearity is half the battle, the other half is concerned with how to get rid of it. Although there are no sure methods, there are only few rules of thumb such as use of extraneous or priori information, omitting highly collinear variable, transforming data and increasing sample size. Serial autocorrelation is an econometric problem which arises when the disturbance terms are correlated to each other. It might be caused due to sluggishness of economic time series, specification bias, data massaging or data transformation. Although it exists, the OLS estimators may remain unbiased, consistent and normally distributed but not efficient. There are formal and informal methods of detecting autocorrelation of which the Durbin-Watson  $d$  test is the popular one. The remedy depends on the nature of the interdependence among the disturbance terms.

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